BAJAMIL Ideas
Findings & Studies

Saudi / GCC marketplace intelligence: data, compliance, business ideas & website feasibility

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This volume is a faithful PDF rendering of the Markdown studies in the repository. No content was changed. Numbers are stated only where the source documents state them; estimates are marked as such in the source. Compliance verdicts are research, not legal advice.

Data Expansion Study — How much more is extractable from the Saudi / GCC fleet?

Date: 2026-11-11 **Scope:** the 12 built clients, the 20-platform candidate fleet (`CANDIDATE_MARKETPLACES_FLEET.md`), and the free "demand signal" sources in `README.md`. **Method:** 8 parallel research subagents, one per independent website cluster, each told to verify endpoints against **primary sources** (official API docs, the site's own JS/SSR payloads, live read-only GETs). Every finding is labelled **VERIFIED** (a source or live response was found) or **INFERRED** (reasoned from platform conventions). No credentials, cookies or tokens were captured or reproduced; no scraping was performed beyond single public read-only probes. This is a feasibility study, not a harvesting run.

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1. Verdict

Yes — there is substantially more data, an estimated 2–4× what the existing clients capture (basis: the public discovery keys already in the Worker expose browse / facet / autocomplete / recommendation / PDP endpoints with several extra fields per SKU on top of today's single search call). But the study's main conclusion is that the constraint has *moved*:

> The project does **not** have an "endpoint discovery" problem. It has a **storage** / **identity** problem. Most of the untapped data sits behind public keys and endpoints the codebase **already holds**, or behind static sitemaps that need no auth at all. Adding a 13th or 30th scraper yields less marginal value than adding a time-series store and a barcode (EAN/GTIN) resolver.

There is also a hard ceiling on the original "as many APIs as possible" ambition: about **10 of the ~34 surfaces examined are partnership-gated** (OAuth contracts, affiliate accounts, seller credentials) and cannot be harvested by a free Worker at all.

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2. Evidence quality — what was actually confirmed live

Concrete probes the agents ran (these are the strongest evidence in the study):

Probe	Result	Consequence
`completion.amazon.sa/api/2017/suggestions?...`	**200, real JSON** (`suggestions[].value`, `responseId`)	Only genuinely free Amazon surface
`gcc.luluhypermarket.com/en-sa/list/?format=json`	**200, 11,843 SKUs / 593 pages**, facets, `ean`, `taxRate`, `isGlutenFree`, `stockCount`	Highest value/effort new dataset
Nahdi `/en-sa/plp/*`	**200, 699KB RSC payload** embedding real Algolia hits + config	Free, no WAF
`api.salla.dev/marketplace/v2/apps`	**200 anonymous JSON**	Zero-friction enrichment
`POST graphql.haraj.com.sa`	**200** `{"data":{"__typename":"Query"}}` anonymous	Open GraphQL
Aqar / Syarah `sitemap.xml` children	**200 static XML**, robots-allowed	Free time-series
`saudi.sharafdg.com/wp-json/wc/store/v1/products`	**403 (WAF)**	Must use Algolia key instead
`al-dawaa.com` homepage + OCC API	**403 Cloudflare "Attention Required!"** even with browser UA	**Contradicts fleet doc** (see §4)

`alsaifgallery.com/graphql` via GET	**404** (POST-only / CDN)	Expected, not a blocker
`reddit.com/r/.../new.json`	**403**	Arctic Shift remains the path
Google Trends `/api/explore` + Books API	**429** / **404**	Exclude from free cron

Platform mechanisms (Magento GraphQL schema, SAP OCC v2, Spree Storefront, Algolia REST, WooCommerce Store API, SFCC controllers, Constructor.io, Unbxd, ZATCA) were verified against **official vendor documentation**; site-specific field presence, index names and base-site IDs remain INFERRED.

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3. The five cross-cutting findings

Finding 1 — The biggest untapped data is behind keys we already hold

The `cloudflare-worker/src/index.ts` already stores public Jarir (Constructor.io), eXtra (Unbxd) and Noon keys. Each vendor's *other* endpoints multiply yield with **zero new credentials and zero new infra**:

- **Jarir / Constructor.io**: `/browse/{facet}/{value}`, `fmt_options.show_hidden_fields` (exposes `inventory_count`, `store_availability`), `variations_map`, `/v1/autocomplete/`, `/v1/recommendations` (best-sellers, bought-together), `lang=ar` bilingual, facet-option counts. VERIFIED `security: []` = no auth on search/browse. Constructor.io docs
- **eXtra / Unbxd**: `/category` (facet counts + `stats=price` min/max + `variants=true`), `/autosuggest` (`topQueries`, `trendingSearches`, `keywordSuggestions` — a **direct demand signal**), `didYouMean`. Unbxd docs
- **Noon**: the undocumented PDP object `/_svc/catalog/api/v3/u/{ZSKU}/p/` returns feature bullets, variants, media, MRP, specs/EAN in one GET — completely missed by the search-only client (TLS fingerprinting makes this PARTIAL/risky from a Worker).

The same pattern repeats on the candidate side: **Tamimi uses Constructor.io** (same stack as Jarir, near-zero marginal engineering), **Danube, Nahdi, Sharaf DG and Sun & Sand Sports expose public Algolia search-only keys**, and **three electronics retailers (Alsaif, Eddy, Black Box) share one unauthenticated Magento2 `POST /graphql` schema** (`custom_attributesV2` spec dictionaries, `aggregations`, related/upsell/crosssell graphs, `sa_ar`/`sa_en` dual store views).

Finding 2 — Sitemaps are the universal free fallback, and they unlock the missing history layer

The earlier 8.7/10 project review named **"no persistent unified data storage / price history"** as gap #1. Three sites leak exactly the raw material for it through robots-allowed static XML:

- **Haraj**: hourly partitioned sitemaps (`sitemapfiles.haraj.com.sa/YYYY-MM-DD/sitemap-*-HH.txt`) plus `postDate`/`updateDate`/`before_update_date` cursors → listing birth, refresh, relist, death events.
- **Aqar**: date-partitioned sale/rent sitemaps **2024-02 → 2026-11** with `<lastmod>`, plus `city/district/filter/price/{id}.xml` hub IDs → cohort, seasonal and price-drop proxies. Detail pages are Cloudflare-403 to cloud IPs, but the sitemaps are not.
- **Syarah**: `cardetail`, `prices`, `make_model_year` sitemaps and a `cars-by-ids` (100 IDs) re-poll → clean per-car price history. `/api/syarah_v1/` is 401 token-gated, sitemaps are open.

This is the cheapest possible route to the project's single biggest structural gap: **no auth, no keys, no WAF, plain `GET` + parse into D1/SQLite.**

Finding 3 — The real ceiling is partnership, not scraping

A clean three-way split emerged. "More APIs" is bounded by credentials, not by cleverness:

Tier	Platforms	Access

Free / anonymous (Worker-viable)	LuLu, Danube, Tamimi, Panda, Nahdi, **Whites (new)**, Alsaif, Eddy, Black Box, Sharaf DG (via Algolia), Namshi, Sun & Sand Sports, Haraj, Aqar/Syarah sitemaps, Sabbar, Kafiil, Salla App Store, Etimad visitor, Mostaql, Khamsat, Amazon completion, all free signal sources	plain HTTP / public keys
Partial / conditional	Carrefour (geo + BFF), Noon PDP (TLS), Virgin OCC (guest token + baseSiteId), OpenSooq (HMAC device handshake), Jahez consumer pages	needs extra posture
Partner-gated (no free path)	Zid (`X-Manager-Token` + OAuth, 60 rpm), Salla merchant/partner APIs, Jahez `integration-api`, Ounass (Partnerize), Boutiqaat (Admitad/ArabClicks), Sivvi / Noon Partner API, Amazon deep catalog, Product Hunt GraphQL, X API	**approved account required**

Evidence caveat: only the 11 live probes in §2 are **VERIFIED** at the HTTP level. Most tier placements are **INFERRED** from vendor docs and platform conventions — the Alsaif/Eddy/Black Box Magento `POST /graphql`, Tamimi's Constructor key and the Algolia index names were not confirmed live. Treat the tier column as a routing hypothesis: run one read-only probe per site before committing engineering time.

Notable: **Amazon PA-API 5 is deprecated (HTTP 403 `AccessDeniedException`)** and the replacement Creators API needs credentials **plus 10 qualifying sales / 30 days** — so the fleet doc's "autocomplete only" recommendation is correct and deep Amazon catalog is not a free target. Amazon Creators API

Finding 4 — Demand signals are a parsing change, not a new integration

The richest untapped signal data is **already inside endpoints the daily cron calls**; it is simply discarded:

- **Google Trends SA RSS**: `ht:approx_traffic` (numeric magnitude, e.g. "2000+"), `pubDate` trend-start, and up to 3 `ht:news_item` links — currently only titles are kept. Daily snapshots → slope + anomaly detection. RSS
- **HN Algolia**: `nbHits` (e.g. 14,469 for "salla"), `points`, `num_comments`, `created_at_i`, `numericFilters` → demand magnitude and time-series slope. HN Algolia API
- **Arctic Shift**: full Reddit objects (`score`, `upvote_ratio`, `num_comments`, **author flair = region/city** S A, `created_utc`, crosspost graph) instead of text-only RSS. Arctic Shift
- **Genuinely new free sources**: GitHub REST search (`total_count` 734 repos for "zatca", stars, topics), Stack Exchange API (`view_count`, `score`, `is_answered`), iTunes-SA review RSS (Arabic review text + `im:rating` sentiment), Wikipedia Pageviews. All no-key, rate-limited.

The two highest-value "full" sources are **excluded** from a free cron: Google Trends `/api/explore` (**unofficial, cookie/token handshake, aggressive 429 — we observed 429 here**, so too fragile for an unattended cron) and X/Twitter (free tier discontinued Feb 2026). pytrends post-mortem

Finding 5 — The bottleneck is identity resolution, not coverage

Every cluster independently surfaced **EAN/GTIN/MPN**: LuLu `ean`/`mainEan`, Carrefour GTIN, Al-Dawaa OCC barcodes, Zid product `barcode`, Magento `custom_attributesV2` MPN/GTIN, Ounass style codes, Namshi `parentSku`. Cross-store price comparison — the project's stated commercial purpose — is impossible without a resolver that links those identifiers. This converges exactly with the earlier review's gap #3 ("universal barcode matching"). **The study's recommendation is therefore to build the resolver and the history store before adding more scrapers.**

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4. Corrections to existing documentation

These are the highest-value outputs of the study, because they prevent building on wrong assumptions:

1. **Al-Dawaa is NOT free-Worker-viable.** `CANDIDATE_MARKETPLACES_FLEET.md` ranks it **#1 (8.2/10, "Yes (OCC v2 REST)")**. Live probes returned **Cloudflare 403 on both homepage and OCC API** from a datacenter IP even with browser UA/Referer/Origin. It needs browser-render/TLS impersonation + an anonymous OAuth2 guest token. **Nahdi and Whites** (Next.js + public Algolia SSR payload, no WAF) are the actual easy pharmacy wins.
2. **"15% VAT" is not universal.** ZATCA zero-rates *SFDA/MoH-classified "qualifying medicines"*, **not** a clean Rx/

OTC split — so a per-SKU derivation is required. The LuLu `?format=json` sample returned `taxRate: 5` (field semantics unclear — Saudi statutory rates are 0% / 15%, so this is likely a legacy or category code, not a percent); do not use it as VAT ground truth. Treat the fleet doc's VAT line as a simplification, not ground truth. ZATCA retail VAT guideline

3. **Amazon PA-API 5 is dead** (403); the fleet doc's autocomplete-only path stands, but any "deep catalog via PA-API" plan must be dropped.
4. **Aqar** / **Syarah** / **OpenSooq** detail APIs are gated (403 / 401 / HMAC handshake), but their sitemaps and robots-disclosed paths are free — the fleet doc understates what is harvestable from them.

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5. Priority roadmap

Tier 1 — near-zero cost, do first (all free, keys already in hand or plain GET) 1. Extend Constructor.io, Unbxd and Noon clients with `browse`/`facets`/`autosuggest`/`recommendations`/PDP calls (roughly 2–4× the fields per SKU, no new infra). 2. Harvest Haraj hourly + Aqar date-partitioned + Syarah `cardetail`/`prices` sitemaps into a time-series store (D1 or SQLite). 3. Parse the discarded numeric fields from Trends RSS, HN Algolia and Arctic Shift in `agent.py`. 4. Add the three Magento2 `/graphql` stores (Alsaif, Eddy, Black Box) with one shared schema adapter, plus Tamimi (Constructor) and Danube/Nahdi/Whites (Algolia).

Tier 2 — moderate effort 5. Etimad **award results** endpoint (winning bidder, all bidder prices, awarded value — **INFERRED** from third-party API specs plus Etimad's public visitor endpoints, not live-probed here). Single highest-leverage new dataset if confirmed. 6. Sabbar salary-band matrix, Kafiil/Mostaql bid-count + budget series, Salla App Store review histograms, Panda weekly leaflets + loyalty pricing, Carrefour geo-scoped branch stock. 7. EAN/GTIN cross-store resolver — the prerequisite for real price comparison.

Tier 3 — partnership required (park, don't engineer) 8. Zid / Salla merchant APIs / Jahez B2B / Noon Partner / Partnerize / Admitad — pursue only if a partner account is obtained. Amazon deep catalog and X/Twitter should be excluded from a free-only cron entirely.

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6. Caveats & boundaries

- **VERIFIED vs INFERRED**: vendor mechanisms are well-sourced; site-specific field presence, Algolia app IDs / index names, SAP `baseSiteId`s and internal REST paths are **INFERRED** and must be confirmed by one live probe before implementation.
- **Terms of use**: Google News RSS is licensed for "personal, non-commercial use"; OpenSooq `robots.txt` disallows `/api/*`; several affiliate programs are login-gated. Respect robots, ToS and rate limits. No cookies, tokens or HAR contents belong in the repo.
- **TLS / WAF reality**: Noon (ClientHello fingerprinting), Al-Dawaa (Cloudflare), Sharaf DG `/wp-json` (403), Aqar (403 to cloud IPs) and Amazon (AWS WAF) will not yield to a plain Worker `fetch`; `curl_cffi`-style browser impersonation or residential proxies are required. That is the one place where "free" has a real cost.
- **Scope**: this is a feasibility/desktop study. It did not run `reverse-api-engineer` browser captures, generate clients, or modify any code. Endpoint names here are *targets*, not confirmed-working integrations.

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7. Key primary sources

Vendor / platform docs — Adobe Commerce GraphQL products · SAP OCC v2 REST (SAP Help, Telco Accelerator reference) · Spree Storefront API · Algolia REST search · WooCommerce Store API · SFCC dw.catalog Product · Constructor.io · Unbxd · Salla docs · Zid docs · Strapi REST

Regulatory / open data — ZATCA VAT retail guideline · SASO energy efficiency labels · REGA real-estate indicator · GASTAT · Saudi open data portal

Signals — Google Trends SA RSS · HN Algolia API · Arctic Shift · GitHub REST search · Stack Exchange API · Etimad developer portal

Saudi & GCC E-Commerce Marketplace Intelligence Fleet (Candidate Registry)

This registry tracks the technical evaluation of 20 high-value e-commerce and retail candidates across Saudi Arabia (KSA) and the GCC, researched in parallel by dedicated AI subagents.

Evaluation Criteria

1. **Search & Discovery API:** Dedicated REST/GraphQL, search provider (Algolia, Constructor, Unbxd, Bloomreach, Elastic), or Nuxt/Next SSR JSON.
2. **Data Richness:** Structured SKU, title (AR/EN), SAR pricing, VAT transparency (15%), discounts, stock/availability, brand, and media assets.
3. **Anti-Bot Posture:** Cloudflare Turnstile/WAF, Akamai Bot Manager, Datadome, or open edge API.
4. **Proxy Dependency:** Can it be harvested using 100% free Cloudflare Workers / direct HTTP, or does it require paid residential proxies?
5. **Integration Score:** Rated 1 to 10 based on ease of integration and data yield.

Candidate Fleet Evaluation Matrix

#	Platform	Primary Vertical	Tech / Search Engine	Anti-Bot Level	Free Worker Viable?	Score / 10	Status
1	**Jahez KSA**	Food & Quick-Commerce (PIK)	REST / OAuth 2.0 B2B (`integration-api.jahez.net`)	High (Cloudflare WAF on app)	Partner: Yes / Public: Hard	**7.4 / 10**	**Evaluated**
2	**Amazon SA**	General Retail, Tech & Books	Amazon A9 / HTML / Open Autocomplete	Very High (AWS WAF / Captcha)	Autocomplete: Yes / Catalog: Proxy	**6.5 / 10**	**Evaluated**
3	**Lulu Hypermarket**	Grocery & Department Store	Akinon Commerce / Elasticsearch (`?format=json`)	Moderate / High (Cloudflare)	Yes (via `?format=json`)	**6.8 / 10**	**Evaluated**
4	**Panda / Panda Click**	Hypermarket & Groceries	Next.js SSR + Strapi CMS + ToDoorstep APIs	Moderate (Cloudflare WAF)	**Yes (Next.js / REST)**	**7.1 / 10**	**Evaluated**
5	**Danube / BinDawood**	Premium Grocery & Imports	Spree Commerce / Algolia Client API	Moderate (Cloudflare)	**Yes (Direct Algolia)**	**7.5 / 10**	**Evaluated**
6	**Carrefour KSA**	Hypermarket & Electronics	MAF Retail BFF / Algolia	High (Akamai + DataDome)	Partial (Requires TLS/BFF)	**6.2 / 10**	**Evaluated**
7	**Tamimi Markets**	Specialty Groceries	ZopSmart Headless REST / GCS Assets	Low / Moderate	**Yes (Direct REST API)**	**7.6 / 10**	**Evaluated**
8	**Virgin Megastore**	Gaming, Tech, Books & Culture	SAP Commerce Cloud (OCC v2) / Solr	High (Akamai Bot Manager)	Direct OCC (with headers)	**6.8 / 10**	**Evaluated**
9	**Sharaf DG KSA**	Electronics & Appliances	Enterprise WooCommerce + Algolia InstantSearch	Moderate (Cloudflare CDN)	**Yes (Algolia Facets / REST)**	**7.4 / 10**	**Evaluated**

10	**Black Box KSA**	Consumer Electronics & TV	Headless Magento 2 / GraphQL	Moderate (Cloudflare WAF)	**Yes (Direct GraphQL)**	**7.0 / 10**	**Evaluated**
11	**Eddy Electronics**	Furniture, Appliances & Tech	Magento 2 GraphQL (`/graphql`)	Moderate (AWS WAF / CloudFront)	Yes (via GraphQL)	**7.8 / 10**	**Evaluated**
12	**Alsaif Gallery**	Kitchenware & Small Appliances	Magento 2 GraphQL (`/graphql`)	Moderate (Cloudflare WAF)	**Yes (Direct GraphQL)**	**8.0 / 10**	**Evaluated**
13	**Nahdi Pharmacy**	Healthcare & Baby Care	Adobe Commerce / **Algolia Client API**	Moderate (AWS WAF)	**Yes (Direct Algolia)**	**6.4 / 10** *(7.8 Search)*	**Evaluated**
14	**Al-Dawaa Pharmacies**	Health, OTC & Dermacosmetics	SAP Commerce Cloud (OCC v2) / Solr	Moderate / High (Geo-WAF)	**Yes (OCC v2 REST)**	**8.2 / 10**	**Evaluated**
15	**Namshi KSA**	Fashion, Apparel & Shoes	REST Gateway (`api.namshi.com`)	Moderate (Cloudflare Bot)	**Yes (REST Gateway)**	**8.0 / 10**	**Evaluated**
16	**Sivvi KSA**	Fashion & Beauty	Noon Catalog API (`noon.partners`)	Moderate (Cloudflare WAF)	Yes (via Noon API)	**7.2 / 10**	**Evaluated**
17	**Ounass KSA**	Luxury Designer & Regional Fashion	Node.js Headless / Algolia / Partnerize	Very High (Cloudflare + Kasada)	Partnerize Feeds: Yes	**7.6 / 10**	**Evaluated**
18	**Sun & Sand Sports**	Sports & Athletic Apparel	Salesforce Commerce Cloud (SFCC) / Algolia	Moderate / High (Akamai/ SFCC)	**Yes (SFCC Search / Algolia)**	**7.8 / 10**	**Evaluated**
19	**Zid Ecosystem**	Saudi Merchant Stores	Zid Multi-Tenant API (`api.zid.sa/v1/`)	Moderate (OAuth2 / Dual Token)	Yes (Partner/ Merchant)	**7.7 / 10**	**Evaluated**
20	**Boutiqaat KSA**	Beauty, Perfume & Influencers	Magento 2 / GraphQL / Affiliate	High (Cloudflare Enterprise)	Affiliate Feeds: Yes	**4.5 / 10**	**Evaluated**

Detailed Platform Profiles

1. Jahez KSA (`jahez.com` / `jahez.net`)

- **Domains:** `jahez.com`, `jahez.net`, `integration-portal.jahez.net`, `integration-api.jahez.net`
- **Architecture:** Dual API: Official B2B Partner API with OAuth 2.0 for POS/ERP (`Foodics`, `Deliverect`, `Odoo`), and mobile client APIs (`api.jahez.net`).
- **Coverage:** Pan-KSA covering all Tier-1 and 50+ Tier-2 cities.
- **PIK Retail:** Extends delivery radius up to 100km for non-food retail (electronics, beauty, gifts).
- **SAR Pricing & Tax:** 15% VAT explicitly segmented; delivery fee SAR 10–20; Jahez Prime gives free delivery on orders >= SAR 35.
- **Anti-Bot & Security:** Cloudflare WAF, geo-fenced coordinates, SSL pinning on mobile app.
- **Recommendation:** Integrate via official B2B integration credentials (`integration@jahez.net`) or POS middleware for 100% reliable real-time menu and order feeds.

2. Zid Ecosystem (`zid.sa` / `mazeed.sa`)

- **Domains:** `zid.sa`, `docs.zid.sa`, `apps.zid.sa`, `mazeed.sa`
- **Scale:** 10,000+ active Saudi/GCC enterprise and high-AOV DTC brands (perfumes, abayas, specialty coffee, luxury retail).
- **Consumer Marketplace:** `mazeed.sa` acts as Zid's unified multi-store consumer marketplace.
- **API Architecture:** RESTful multi-tenant API at `https://api.zid.sa/v1/` with OAuth 2.0 Authorization Code grant.
- **Authorization Headers:** Unique dual-token requirement: `Authorization: Bearer <access_token>` plus `X-Manager-Token: <manager_token>`.
- **Rate Limits:** 60 requests/minute per store (Leaky Bucket algorithm). Webhook-first architecture (`order.create`, `product.update`, `inventory.update`) is mandatory.
- **Tax & ZATCA:** Explicit 15% KSA VAT, HS code support, and Phase-2 e-invoicing compliance.
- **Recommendation:** Integrate as a registered App Market public app (`apps.zid.sa`) to unlock all 10,000+ merchants without requiring store-level Pro/Enterprise custom API surcharges.

3. Amazon Saudi Arabia (`amazon.sa`)

- **Domain:** `https://www.amazon.sa` (Marketplace ID: `A17E79C6D8DWNP`)
- **Open Autocomplete API:** Completely unauthenticated live keyword and category recommendation API at `https://completion.amazon.sa/api/2017/suggestions?mid=A17E79C6D8DWNP&alias=aps&prefix={query}&lop=en_AE`.
- **Search Architecture:** `/s?k={query}&page={page}&s={sort}&rh={filters}` with ASIN extraction via `div[data-component-type="s-search-result"]`.
- **Pricing & Locale:** Strict SAR pricing (`lc-main=en_AE; i18n-prefs=SAR;`).
- **Anti-Bot Defense:** Heavy AWS WAF / CloudFront bot detection and CAPTCHA walls on raw datacenter IPs.
- **Recommendation:** Use open Autocomplete API for free zero-proxy keyword intelligence, and route deep catalog queries through managed SERP/proxy scraping.

4. Carrefour Saudi Arabia (`carrefourksa.com`)

- **Operator:** Majid Al Futtaim (MAF) Retail (territory code `mafsau`).
- **API Architecture:** High-performance REST Backend-For-Frontend (`https://www.carrefourksa.com/api/v4/search` and `/api/v8/categories/{id}`).
- **Search Provider:** Algolia Cloud cluster powering instant search suggestions and facet filtering behind MAF BFF.
- **Pricing & Tax:** Native SAR pricing with 15% VAT included.
- **Inventory Dependency:** Strict geolocation requirements (`latitude`, `longitude`, and `areaCode` e.g., Riyadh, Jeddah) for accurate grocery stock.
- **Anti-Bot Defense:** Akamai Bot Manager + DataDome. Requires browser TLS fingerprints (`curl-cffi`) and residential/GCC IPs.

5. Virgin Megastore Saudi Arabia (`virginmegastore.sa`)

- **Platform:** SAP Commerce Cloud (Hybris CCv2) + Apache Solr + Bloomreach Discovery.
- **API Endpoints:** Standard SAP OCC v2 REST endpoints (`/occ/v2/{baseSiteId}/products/search` and `/occ/v2/{baseSiteId}/products/suggestions`).
- **Key Categories:** Gaming (PS5, Xbox, PC Gaming, Razer, Logitech), Books (Arabic/English, Manga), Gadgets/Audio (Sony, Bose, Marshall, Funko Pop).
- **Tax & Pricing:** Standard SAR with mandatory 15% VAT; Mada, Apple Pay, Tabby, and Tamara.
- **Anti-Bot Defense:** Guarded by Akamai Bot Manager / Edge Defender. Direct OCC querying requires browser TLS impersonation and valid guest session tokens.

6. Eddy Home & Electronics (`eddy.com.sa`)

- **Platform:** Adobe Commerce (Magento 2 Enterprise) on AWS multi-AZ, managed by SnapTec.
- **Direct GraphQL API:** `https://eddy.com.sa/graphql` (Supports standard `products(...)` query with full JSON specs, prices, and stock).
- **REST & Autocomplete:** `/rest/V1/` and `/catalogsearch/ajax/suggest`.
- **Catalog Pillars:** Major Appliances (Refrigerators, Washers, ACs with SASO ratings), Consumer Electronics (Smart TVs, Mobiles, Computing, Gaming), and Furniture (Living, Bedroom, Dining).
- **Financing:** Tabby, Tamara BNPL, and 0% bank installment plans across major Saudi banks.
- **Recommendation:** Highest feasibility for consumer tech and appliances; query the `/graphql` endpoint directly.

7. Ounass Luxury Saudi Arabia (`saudi.ounass.com`)

- **Operator:** Al Tayer Digital (Dubai / Riyadh).
- **Catalog:** Tier-1 luxury houses (Gucci, Prada, Saint Laurent, Balenciaga), niche perfumery (Amouage, Parfums de Marly), and regional Saudi designers (modest fashion, abayas, luxury Arabic sandals).
- **Pricing:** SAR with 15% ZATCA VAT and 100% DDP (Delivered Duty Paid) cross-border customs coverage.
- **Integration Routes:** Primary recommended route via official **Partnerize product data feeds (CSV, XML, JSON)** with 30-day attribution. Raw scraping is heavily restricted by Cloudflare Enterprise + Kasada.

8. Boutiqaat KSA / GCC (`boutiqaat.com`)

- **Model:** Influencer-led social e-commerce featuring 700+ celebrity digital boutiques (Fouz Al Fahad, Noha Nabil, Dr. Kholoud).
- **Catalog Strengths:** Rare Gulf perfumery (Gissah, Atyab Al Marshoud, Arabian Oud), international niche beauty, and cosmetics.
- **Platform:** Magento 2 Enterprise + Next.js + Algolia search.
- **Anti-Bot:** Cloudflare Enterprise WAF.
- **Integration Route:** Closed 1P supplier model; external catalog feeds best consumed via regional affiliate networks (Admitad / ArabClicks).

9. Danube KSA / BinDawood (`danube.sa`)

- **Operator:** BinDawood Holding (Tadawul: 4161) / IACo.
- **Platform:** Spree Commerce (Ruby on Rails) + LitElement Web Components PWA.
- **Direct Search API: Algolia** (`algoliasearch` with public Search-Only client key).
- **Catalog Data Yield:** Product ID, barcode/EAN, bilingual titles (`name_en`, `name_ar`), taxons, prices in SAR, 15% VAT, and Amazon CloudFront packshot images.
- **Store Localization:** Inventory availability is tied to local branch / city coordinates (`supermarket_id`).
- **Recommendation: Grade A (9.0/10 for Catalog & Search Discovery)**. Direct Algolia queries completely bypass web scraping barriers and Cloudflare WAF with zero proxy cost.

10. Alsaif Gallery KSA (`alsaifgallery.com`)

- **Company:** Leading Saudi listed houseware retailer (Tadawul: 4192) with 60+ megastores.
- **Platform:** Decoupled **Adobe Commerce (Magento 2)** + Headless Next.js (NOT Salla!).
- **Direct GraphQL Endpoint:** `POST https://alsaifgallery.com/graphql` (Store: `sa_ar` or `sa_en`).
- **Flagship Brand: Edison (إديسون)** appliances (pressure cookers, air fryers, stand mixers, Arabic bread bakers).
- **Pricing & Offers:** 15% VAT included, regular vs final discount prices, Tabby, Tamara BNPL.
- **Recommendation: High Feasibility (8.0/10)**. Querying the `/graphql` endpoint yields clean JSON without parsing messy HTML.

11. Namshi Saudi Arabia (`saudi.namshi.com`)

- **Operator:** Noon Group (acquired for $335M).
- **Scale:** 100,000+ active fashion SKUs, 2,000+ brands (Nike, Mango, Aldo, Ella, Amouage).
- **REST Gateway:** `https://api.namshi.com/catalog/search?q={query}` and `/catalog/products/{sku}`.
- **Mandatory Headers:** `X-Country: SA`, `X-Language: en|ar`, mobile `User-Agent`.
- **Media & Logistics:** Media hosted on Noon CDN (`f.nooncdn.com`), fulfillment via Noon Express.
- **Pricing & Tax:** 15% ZATCA VAT compliant gross pricing in SAR, Tabby/Tamara BNPL.
- **Recommendation: Grade A- (8.0/10)**. Standard REST JSON endpoints are highly accessible via mobile client header emulation.

12. Sivvi KSA (`sivvi.com/saudi-en`)

- **Platform:** Fast-fashion vertical of Noon AD Holdings.
- **Integration Channel:** Standardized under **Noon Partner API** (`noon.partners` / `api.noon.com` with `Contract: SA`).
- **Catalog:** High-street fashion, athleisure, and cosmetics sharing Noon PIM catalog taxonomy.
- **Fulfillment:** Same-day and sub-3-hour delivery in Riyadh via Noon Express & Noon Minutes.
- **Recommendation:** Integrate via Noon Partner API alongside Noon marketplace catalog feeds.

13. Black Box KSA (`blackbox.com.sa`)

- **Company:** Saudi omnichannel white-goods and consumer electronics retailer.
- **Platform:** Headless Adobe Commerce (Magento 2) + Next.js frontend (`store.ops.blackbox.com.sa`).
- **Direct API:** Magento GraphQL endpoint (`https://blackbox.com.sa/graphql`) and REST `/rest/V1/products`.
- **Catalog Coverage:** Daikin, Gree, LG, Samsung split ACs, home theater OLEDs, laptops, and refrigerators.
- **Pricing & Tax:** 15% ZATCA VAT included in SAR, Tabby and Tamara 4-split BNPL.
- **Recommendation: Feasible (7.0/10)**. Intercepting GraphQL queries yields structured JSON with specs and stock status.

14. Nahdi Pharmacy Online (`nahdionline.com`)

- **Company:** Saudi Arabia's #1 pharmacy chain (1,100+ stores).
- **Search Engine: Algolia DSN** (`window.algoliaConfig` with public search-only API keys).
- **Direct Search API:** `https://{APPLICATION_ID}-dsn.algolia.net/1/indexes/{INDEX_NAME}/queries`.
- **Catalog:** 25,000–35,000 SKUs (OTC analgesics, diabetes monitors, vitamins, baby formula, dermacosmetics).
- **Dual VAT Structure:** SFDA regulated drugs (0% VAT, fixed pricing), OTC & wellness (15% VAT included).
- **Recommendation: Grade A (7.8/10 for Catalog & Search Aggregation)**. Direct Algolia queries bypass AWS WAF and CloudFront with sub-50ms latency.

15. Al-Dawaa Pharmacies KSA (`al-dawaa.com`)

- **Company:** Tadawul: 4163. Over 975+ pharmacies, 23+ dark stores across KSA. FY25 Revenue: SAR 6.74 Billion.
- **Platform: SAP Commerce Cloud (Hybris)** + Headless Angular Spartacus SPA.
- **Core API:** SAP Omni Commerce Connect (**OCC v2 REST API**) under `https://api.al-dawaa.com/occ/v2/aldawaa/products/search`.
- **Search Engine: Apache Solr** with bilingual Arabic/English analyzers and SFDA medication code indexing.
- **Catalog Depth:** 25,000–35,000 SKUs covering healthcare, French dermacosmetics (La Roche-Posay, CeraVe, Vichy), infant nutrition, and diagnostic devices.
- **Recommendation: Grade A- (8.2/10)**. High-fidelity SAP OCC v2 REST endpoints return clean JSON with EAN barcodes and live store availability.

16. LuLu Hypermarket KSA (`luluhypermarket.com/en-sa`)

- **Platform:** Headless React SPA powered by **Akinon Commerce Cloud (ACC)** + **SAP Commerce Cloud (Hybris)**.
- **Direct JSON Flag:** `GET https://www.luluhypermarket.com/en-sa/list/?search_text={q}&format=json` (returns structured JSON containing pagination, facet aggregations, EAN barcodes, and product objects).
- **Search & Autocomplete:** Elasticsearch via Akinon (`/en-sa/autocomplete/?search_text={prefix}`).
- **URL Hierarchy:** Department URLs (`/en-sa/{category-slug}/c/{categoryCode}`) and PDP URLs (`/p/{sku-code}/`).
- **Pricing & Tax:** 15% VAT included in SAR (`vatRate: 15.0`), retail vs selling discount prices.
- **Recommendation: Solid B+ (6.8/10)**. Clean Akinon `format=json` catalog payloads make extraction fast when paired with Cloudflare TLS headers.

17. Sun & Sand Sports KSA (`en-sa.sssports.com`)

- **Operator:** Gulf Marketing Group (GMG). Largest sports retailer in the Middle East.
- **Platform: Salesforce Commerce Cloud (SFCC / B2C Commerce)** + Storefront Reference Architecture (SFRA).
- **Search Engine: Algolia AI Search** integrated client-side with SFCC for typeahead autocompletion, instant search, and facet filtering.
- **Core Endpoints:**
- SFCC Search Controller: `https://en-sa.sssports.com/on/demandware.store/Sites-en-sa-Site/default/Search-Show?q={query}`
- Product Detail Controller: `https://en-sa.sssports.com/on/demandware.store/Sites-en-sa-Site/default/Product-Show?pid={pid}`
- Suggestions: `Search-GetSuggestions`
- **Enterprise Layer:** Akeneo PIM for multilingual product attributes, Fluent Commerce for OMS / click & collect, SAP S/4HANA ERP.
- **Tax & Currency:** All prices in SAR with 15% ZATCA VAT included. Supported by Tabby/Tamara BNPL, Apple Pay, Mada.
- **Anti-Bot & Infrastructure:** Akamai Edge Defender / SFCC DDoS shield. Product pages follow clean slug patterns: `en-sa.sssports.com/<slug>/<sku>.html`.
- **Recommendation: High Feasibility (7.8/10)**. SFCC search controllers and Algolia client requests can be queried directly with standard browser headers.

18. Panda / Panda Click KSA (`panda.sa` / `panda.com.sa`)

- **Operator:** Panda Retail Company (Savola Group). Largest grocery & hypermarket network in Saudi Arabia (200+ stores).
- **Frontend Architecture: Next.js (React)** with Server-Side Rendering (SSR) under `https://panda.sa/en/` and `https://panda.sa/ar/`.
- **Backend & Mobile Ecosystem:** Headless microservices (Node.js/NestJS), Strapi CMS for marketing, and legacy **ToDoorstep** app fulfillment APIs (`todoorstep.com` / `com.todoorstep.store`).
- **CDN & Asset Pipeline:**
- Product Packshots: `https://images.todoorstep.com/product/{SKU_ID}/En.jpg` and `/Ar.jpg`
- Next.js Image Optimization: `_next/image`
- **Coverage:** Nationwide grocery delivery spanning 20+ Saudi cities (Riyadh, Jeddah, Makkah, Madinah, Dammam, Khobar, Tabuk, Abha).
- **Pricing & Tax:** Strict SAR pricing with 15% VAT included. Weekly flyer promos and bundle discounts updated every Wednesday.
- **Recommendation: Grade B+ (7.1/10)**. Can be queried via Next.js SSR hydration data props (`__NEXT_DATA__`) or by directly querying the ToDoorstep catalog CDN by SKU.

19. Tamimi Markets KSA (`shop.tamimimarkets.com`)

- **Operator:** Tamimi Group. Saudi Arabia's premier upscale fresh foods and imported supermarket chain.
- **E-Commerce Platform: ZopSmart (SmartStore)** omnichannel retail engine.
- **REST Catalog API:**
- Storefront Catalog: `GET https://shop.tamimimarkets.com/api/product?category={catId}&layoutType=grid`
- Open Auth Service: OpenAPI / Swagger endpoints under `/api/v2/Auth/` (e.g. `RegisterAnonymousApp`, `TokenApp`)
- Catalog Asset CDN: Google Cloud Storage bucket `https://storage.googleapis.com/tm-zopsmart-uploads/`
- **App Ecosystem:** White-label mobile app package `com.zopsmart.scarlet` (iOS / Android) talking directly to the ZopSmart REST API.
- **Catalog Highlights:** Premium imported groceries (Safeway Signature Select, O Organics, Kirkland), organic produce, gluten-free pantry, and fresh butchery.
- **Tax & Pricing:** Standard SAR with 15% VAT itemized; Mada, Apple Pay, Credit Cards.
- **Recommendation: Grade A- (7.6/10)**. Clean, structured ZopSmart REST JSON endpoints with minimal anti-bot friction, making it one of the easiest grocery platforms to query programmatically without residential proxies.

20. Sharaf DG Saudi Arabia (`saudi.sharafdg.com`)

- **Operator:** Sharaf Group. Premier regional consumer electronics and technology retailer.
- **Architecture:** Enterprise-tailored **WooCommerce** / **WordPress** frontend paired with **SAP Commerce Cloud (Hybris)** backend order management (`com.sharafdg.oms`) and Openbravo retail POS.
- **Search & Facets Engine: Algolia InstantSearch** powering instant query autocompletion, real-time facet filtering (`dFR[...]`), and search results.
- **Catalog & CDN:**
- Product Categories: `https://saudi.sharafdg.com/c/{category_slug}/` (e.g. `mobiles_tablets`, `computing`, `home_appliances`, `tv_video_audio`)
- Direct Product URLs: `https://saudi.sharafdg.com/product/{product-slug}/`
- Image Delivery: Dedicated CDN `s.sdgcdn.com` and Cloudflare edge image optimizer (`pimcdn.sharafdg.com/cdn-cgi/image/...`)
- **Pricing & Tax:** 15% ZATCA VAT included in SAR, Apple Pay, Mada, Tabby/Tamara installments.
- **Recommendation: High Feasibility (7.4/10)**. Structured category taxonomy and public Algolia InstantSearch integration allow fast, automated product and price extraction.

Executive Summary & Implementation Recommendations

Top 5 Easiest Platforms to Add (Zero-Proxy / Cloudflare Worker Viable)

1. **Al-Dawaa Pharmacies (`al-dawaa.com`) \u2014 Score 8.2 / 10**: Standard SAP OCC v2 REST API (`/occ/v2/aldawaa/products/search`) returning pristine JSON with barcodes and stock.
2. **Namshi KSA (`saudi.namshi.com`) \u2014 Score 8.0 / 10**: Dedicated REST gateway (`api.namshi.com/catalog/search`) with mobile user-agent header.
3. **Alsaif Gallery (`alsaifgallery.com`) \u2014 Score 8.0 / 10**: Direct Magento 2 `POST /graphql` endpoint returning clean product schemas.
4. **Sun & Sand Sports (`en-sa.sssports.com`) \u2014 Score 7.8 / 10**: Salesforce Commerce Cloud `Search-Show` controller + client-side Algolia.
5. **Tamimi Markets (`shop.tamimimarkets.com`) \u2014 Score 7.6 / 10**: ZopSmart headless `/api/product` REST endpoints with no aggressive bot walls.

Top Algolia Client Key Exploits (Instant Sub-50ms JSON)

- **Danube KSA (`danube.sa`)**: Exposes public Search-Only Algolia client key (`algoliasearch`), completely

bypassing Spree Commerce web pages.

- **Nahdi Pharmacy (`nahdionline.com`)**: Client-side `Nahdi_Algolia` DSN endpoint queries live inventory and SFDA medications with zero Cloudflare challenge.
- **Sharaf DG KSA (`saudi.sharafdg.com`)**: Algolia InstantSearch query parameterization.

Top GraphQL Endpoints (Single-Query Spec & Price Extraction)

- **Eddy Electronics (`eddy.com.sa/graphql`)**
- **Alsaif Gallery (`alsaifgallery.com/graphql`)**
- **Black Box KSA (`blackbox.com.sa/graphql`)**

Consolidated Study — State of `BAJAMIL/ideas` after the reliability & compliance work

Date: 2026-11-11 **Method:** full verification sweep, then 3 parallel isolated research subagents on independent domains (coverage-vs-compliance, data yield, reliability/provenance), each told to ground every claim in repo file:line evidence. The coordinating agent independently re-verified the highest-stakes claims: the "12 vs 13" count, the published reports' missing health blocks, and the fleet-score arithmetic. **Inputs (primary sources, all in-repo):** `README.md`, `agent.py`, `marketplace_registry.py`, the 13 `*_client.py` files, `signal_health.py`, `config.py`, `tests/`, `LICENSE`, `pytest.ini`, `.github/workflows/daily-ideas.yml`, `cloudflare-worker/src/index.ts`, `DATA_EXPANSION_STUDY.md`, `CANDIDATE_MARKETPLACES_FLEET.md`, `docs/TOS_AUDIT.md`, `LATEST_IDEAS.md`, `reports/2026-11-10.md`.

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0. Verification sweep (fresh, before any study)

Check	Result	Exit
`python -m pytest`	**27 passed in 0.27s**	0
`python -c "import agent, config, signal_health, marketplace_registry"`	OK, `GOOGLE_NEWS_RSS_ENABLED = False`	0
`python -m compileall` (excl. tests/docs)	clean	0
`git diff --check`	clean (LF→CRLF warnings only)	0
All 8 prior deliverables present	LICENSE, TOS_AUDIT, study, plan, tests, config, pytest.ini, signal_health	—
HEAD / branch	`8cb1082` on `main`	—
Working tree	4 modified, 3 new untracked (+ pre-existing `rtx4070_report.json`)	—

Bottom line: the plumbing is green. The rest of this study is about whether the *story* built on that plumbing holds up — and it partly does not.

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1. The central finding: coverage grew, identity did not

The repo now has **13 marketplace clients** (LuLu was added since the earlier audit), 9 signal collectors, and a 3-store Cloudflare Worker. Yet the three independent studies converge on one conclusion:

> **The project's stated commercial purpose — cross-store price comparison in SAR — is not achievable today.**

Evidence from the field inventory (all 13 clients read from their dataclasses / return shapes):

- **No shared product identity.** Clients emit store-local `sku`/`id`. Only Jarir carries any identifier (`isbn` for books, `item_number` from `bar_code1` metadata — `jarir_client.py:295,302`). LuLu's *source* JSON contains `ean`/`taxRate` (per `DATA_EXPANSION_STUDY.md:26`) but `lulu_client.py:85-90` **drops them**. No client emits a shared EAN/GTIN key; the Worker's inline mappings (`index.ts:154-170,191-200,227-235`) use per-store `price_sar` + `sku`, again with no shared key.
- **Inconsistent money typing.** `int` (Aqar), `float` (Jarir/eXtra/LuLu/NiceOne/GoldenScent), raw strings (Haraj/OpenSooq/Kafiil/Sabbar). Only eXtra and LuLu expose a stock field; only Jarir/eXtra are bilingual.
- **The `/api/compare` endpoint isn't a comparison.** `index.ts:67-99` juxtaposes three separate result lists under

`retailers.jarir/extra/noon`; it never matches the same product across stores.

The single missing primitive is a cross-store EAN/GTIN identity resolver (barcode/MPN → canonical product key). This is exactly `DATA_EXPANSION_STUDY.md` Finding 5 (`:84-85`) and Tier-2 item 7 (`:111`) — still not started. A 14th scraper adds another incomparable price list, not a comparison.

Tier-1 status (from `DATA_EXPANSION_STUDY.md:103-106`)

Item	State	Evidence
LuLu free-Worker dataset	**DONE** (added since the study)	`lulu_client.py`, wired `marketplace_registry.py:33,52,217-229`
Signal health / provenance guard	**DONE**	`signal_health.py`, `marketplace_registry.py:20`
Persistence / time-series store	**NOT DONE**	Worker `Env` empty (`index.ts:12-14`); `haraj_market_engine.py` SQLite is standalone, not imported by registry
EAN/GTIN resolver	**NOT DONE**	no module
Constructor/Unbxd browse + autosuggest extensions	**NOT DONE**	Worker has only `/api/jarir`, `/api/compare`, `/api/jarir/categories`; `browse_category` just wraps `search_products` (`jarir_client.py:497`)
Sitemap harvest	**NOT DONE**	no sitemap code in any `.py`

Fleet conversion (arithmetic re-verified)

Of the 20 platforms scored in `CANDIDATE_MARKETPLACES_FLEET.md`, **exactly 1 is built** (LuLu, 6.8/10). The other 12 clients aren't among the 20 candidates; the 19 fleet names have no client.

- Built (1): sum 6.8 → avg **6.80**
- Unbuilt (19): sum **136.7** → avg **7.195**
- All 20: sum **143.5** → avg **7.175**

So the one converted platform scores **below** the unbuilt average, and the fleet doc's "Top 5 Easiest" (Al-Dawaa 8.2, Namshi 8.0, Alsaif 8.0, S&S 7.8, Tamimi 7.6 — `:223-228`) remain entirely unbuilt. *(Caveat: `DATA_EXPANSION_STUDY.md:93` downgrades Al-Dawaa to non-viable after a live 403.)* This quantifies the pattern: **past effort went to breadth, not to the identity/persistence primitives the engine needs.**

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2. Coverage vs compliance: ~22 sources, zero commercial licences

Counting distinct wired sources: **9** signal collectors in `agent.py`, **13** registry clients, **4** Worker hosts → **~22 distinct** after de-duplication (VERIFIED line references; de-dup logic INFERRED). Using only the verdicts in `docs/TOS_AUDIT.md`:

Bucket	Count	Sources
HIGH (robots-disallowed / personal-only)	**7**	Google Autocomplete, Reddit RSS, Arctic Shift, OpenSooq, Aqar, Nice One, Golden Scent
Gated / already handled	**1**	Google News RSS (`GOOGLE_NEWS_RSS_ENABLED`, default `False`)
M (unclear)	**11**	Trends, Mostaql, Khamsat, Product Hunt, Haraj, Jarir, eXtra, Etimad, Noon, LuLu, Jarir-Nuxt
L / L–M (unclear, lower friction)	**5**	HN Algolia, Salla, Syarah, Sabbar, Kafiil

Zero sources are declared "compliant" (`docs/TOS_AUDIT.md` §2, §4; `README.md:77`). Only 1 of ~22 is actually

gated, while §4 names ~10 more that should be default-OFF.

If the audit's own advice were enforced: §4's "Lowest-risk to keep" names 6 (HN Algolia, Sabbar, Syarah, Kafiil, Salla, Haraj). That leaves **~6 of 22 ≈ 27%** of the daily pipeline, and the registry would drop from 13 to 6 clients (**6/13 ≈ 46%**). Every one of those six is still `UNCLEAR`, not licensed.

Top 5 contradictions (exact quotes + file:line, all VERIFIED)

1. `README.md:3,14` **"100% free APIs"** / **"100% Free Data Sources Queried Every Run"** vs audit §2: `UNCLEAR ≠ compliant`; §4: "No source is declared 'compliant'".
2. `README.md:15` **"completely bypassing Reddit's 403 rate-limiting walls"** and `agent.py:112` **"Bypasses Reddit's strict 403 API blocking"** vs `www.reddit.com/robots.txt` `User-agent: * Disallow: /` + Data API Terms §3.1/3.2 — audit §3 calls it "**Clearest breach**".
3. `README.md:31,32` and `LATEST_IDEAS.md:7,8,38,39` list **Google Autocomplete** / **Trends** as live free sources; audit §3 says they have "the same structural gap that got Google News disabled" and §4 recommends gating both.
4. `marketplace_registry.py:2` **"Registry (12 Platforms)"** while `:33,52` wire a **13th** (`LuluClient`). Confirmed: 13 `self.* = *Client()` instantiations, and **8 places** across `agent.py`, `marketplace_registry.py`, `signal_health.py`, `tests/test_signal_health.py` still say "12 platforms" (re-verified by grep).
5. `collect_market_intelligence()` gates **only** Google News (`agent.py:500`) yet calls Reddit/Autocomplete/Trends/HN/ProductHunt/registry unconditionally (`:487–529`); `config.py` has exactly one flag. `README.md:75` presents "disabled by default" as if the compliance problem were solved.

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3. The verification story is honest about *plumbing*, not about *truth*

The reliability study re-ran every gate (all PASS) and confirmed `build_prompt()` genuinely injects `INTEGRITY_RULES` (`agent.py:45,544`) and that the health block marks failures UNAVAILABLE (`signal_health.py:44–58`). But it found a sharp gap:

- **Both published reports predate the guardrail.** `reports/2026-11-10.md` and `LATEST_IDEAS.md` (commits `20bf4ad`, 12:15) were generated before the guardrail commit (`edec0b8`, 12:27). Re-verified: all three published reports contain **0** `UNAVAILABLE` tokens and **0** `Source availability` blocks.
- **They are saturated with unsourced numeric claims** that the new `INTEGRITY_RULES` explicitly forbid:
- `reports/2026-11-10.md:14` `20,000) "الطقس غدًا"+ daily searches)` — Google Trends RSS yields titles, autocomplete yields strings; neither yields absolute volumes.
- `:16` `"cuaca besok" (expat influence)` — the exact behaviour `INTEGRITY_RULES:20` was written to forbid.
- `LATEST_IDEAS.md:17` 63) أكثر من 63% من سكان المملكة دون سن الثلاثين% under 30) — no collector returns census data.
- `:20,91,323` `300%–600%` margins; `:84` conversion rate `>40%`; `:83` "channel #1" — no source section carries these.
- `README.md:27` `290,000+ Saudi ministry tenders` — no fixture asserts it.
- Verdict: **every audited numeric/live claim is likely-inferred or unverifiable**, not literal-source-plausible.

What the 27 tests do NOT cover

- **Live network behaviour** — every collector is stubbed (`tests/test_tos.py:12`, `test_agent.py:19–26`, `test_registry.py:9–13`).
- **The 13 clients' real responses** — `test_registry.py` injects a fabricated dict; no test asserts any client returns real data.
- **The LLM generation step** — `generate_with_openrouter`, `run_daily_discovery`, the Antigravity branch are untested.
- **GitHub Actions cron** / **deploy**, **Cloudflare Worker build**, **robots.txt enforcement**, and the `is_available()` head-only (first 200 chars) heuristic.

So **"verified" currently means "the wiring is internally consistent", not "the output is true"** — the guardrail

protects *future* runs only.

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4. Synthesis: what all three studies agree on

#	Conclusion	Supported by
1	**Stop adding scrapers.** The binding constraint is identity + persistence, not coverage.	data-yield study; `DATA_EXPANSION_STUDY.md:13,85,111`
2	**The commercial promise is unfulfilled today** — `/api/compare` is a side-by-side list, not a comparison.	field inventory; `index.ts:67-99`
3	**Compliance is documented, not enforced** — ~22 sources, 0 licensed, 1 gated, ~10 more recommended for gating.	compliance study; `docs/TOS_AUDIT.md` §4
4	**Provenance is unproven** — published reports contain unsourced numbers; the guardrail only applies to future runs.	reliability study; re-verified token counts
5	**Docs drift is measurable** — "12 platforms" in 8 places, "100% free", "bypass 403", a 13th client added without a docstring update.	all three studies; grep re-verification

Recommended order (derived, not started)

1. **Extract LuLu's `ean` into a canonical key** → build the EAN/GTIN resolver → then persistence/time-series. These two unlock the actual product. *(Highest value, still unstarted.)*
2. **Regenerate or annotate the published reports** with a Source availability block, or clearly label them pre-guardrail — otherwise the README describes behaviour the shipped reports don't exhibit.
3. **Fix the 8 stale "12 platforms" strings** and the README's "100% free" / "bypass 403" framing.
4. **Enforce the audit's gates** (`docs/TOS_AUDIT.md` §4) — this is a *separate bounded change* not yet implemented; note it would cut the daily pipeline to ~27%.
5. **Add live smoke tests** for at least the lowest-risk clients (HN Algolia, Sabbar, Kafiil), so "verified" starts to cover real data.

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5. Limitations & boundaries

- Compliance verdicts are taken **verbatim** from `docs/TOS_AUDIT.md` (itself research, not legal advice); no new web ToS research was done in this pass. `UNCLEAR` does not mean compliant.
- The "~22 sources" and the "~27% / 46%" reductions involve de-duplication and denominator choices that are **INFERRED**; the per-source line references are VERIFIED.
- The fleet arithmetic was recomputed independently (built 6.80 / unbuilt 7.195 / all 7.175) and matches the subagent's figures up to rounding.
- The numeric-claim verdicts are based on **absence** of matching source sections plus the guardrail text — a plausible inference, not proof of fabrication (the LLM may have used general knowledge).
- This study made **no code changes, no commits, no pushes**. Working-tree changes from earlier turns remain uncommitted.
- Live endpoint behaviour (WAF blocks, 403s, current `robots.txt`) was not re-probed here; see `DATA_EXPANSION_STUDY.md:25-37` for the earlier live probes.

Third-Party Source ToS / `robots.txt` Audit

Date: 2026-11-11 **Scope:** all data sources wired into `agent.py`, the 12 `*_client.py` marketplace clients, and the Cloudflare Worker's public discovery keys. **Method:** one isolated research subagent audited each source against primary terms / `robots.txt` files; four highest-stakes claims were independently re-fetched and confirmed by the coordinating agent (Reddit, Google Trends, Golden Scent, Nice One). **Legal posture assessed: commercial/ business use** — `SYSTEM_PROMPT` designs priced products (3x–6x markup, 19–79 SAR, Grand Slam Offers, Tabby/ Tamara) and the output is published to GitHub Pages + a GitHub Issue. A side project with no revenue is still assessed as business use.

> ⚠ **This is research, not legal advice.** `VERIFIED` means a primary source was found and read; `UNCLEAR` means no explicit prohibition *or* permission was found — it does **not** mean "compliant". Confirm with counsel before commercial deployment.

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1. Already handled (not re-audited)

Source	Status
Google News RSS	Verified against Google News Terms of Use §2: *"personal use (i.e., non-commercial use)"*, *"may not use the Service to sell a product or service"*. Now **disabled by default** behind `GOOGLE_NEWS_RSS_ENABLED` in `config.py`; when off, no request is made and the source is not listed in the health block.

2. Findings table

Source	Endpoint used	Governing terms	Commercial use?	`robots.txt` disallow?	Risk
Google Autocomplete	`suggestqueries.google.com/complete/search?client=firefox&gl=sa&hl=ar&q=…`	Google ToS	**UNCLEAR (lean NO)**	Host returns 404 (no robots); `www.google.com/robots.txt` disallows `/search`	**H**
Google Trends RSS	`trends.google.com/trending/rss?geo=SA`	Google ToS	**UNCLEAR**	`trends.google.com/robots.txt` disallows only `/explore?` and `/trends/explore?`; **RSS path not disallowed** (re-fetched, confirmed)	**M**

Reddit RSS (`.rss`, `search.rss`)	`www.reddit.com/r/{sub}/search.rss`, `/new.rss`	Reddit robots.txt → Public Content Policy; Data API Terms	**PERSONAL-ONLY / NO**	**`User-agent: *` `Disallow: /`** (all paths) — re-fetched, confirmed	**H**
Arctic Shift (Reddit fallback)	`arctic-shift.photon-reddit.com/api/posts/search`	No published ToS found; upstream Reddit content still governed by the above	**UNCLEAR**	n/a (API host)	**H**
Mostaql	`mostaql.com/projects` (HTML regex)	Terms (Arabic)	**UNCLEAR (lean NO)**	`Disallow: /search*`, `/ajax/`; `/projects` **not** disallowed	**M**
Khamsat	`khamsat.com/community/requests` (HTML regex)	Terms (Arabic)	**UNCLEAR (lean NO)**	`Disallow: /search*`, `/community/search*`, `/ajax/*`; requests board **not** disallowed	**M**

Hacker News Algolia	`hn.algolia.com/api/v1/search_by_date`	HN Algolia API	**UNCLEAR (lean YES)**	`/robots.txt` 404 (none)	**L**
Product Hunt feed	`www.producthunt.com/feed`	Legal (403 on fetch)	**UNCLEAR (lean NO)**	`/feed` **not** disallowed; `/search*` is	**M**
Haraj	`graphql.haraj.com.sa`, `matjar.haraj.com.sa/graphql`	No public API terms found	**UNCLEAR**	`haraj.com.sa/robots.txt`: `Allow: /`; gateway is a separate host	**M**
OpenSooq	`sa.opensooq.com` + Next.js SSR state	No English ToS found	**UNCLEAR (lean NO)**	**`Disallow: /api/*`, `/_next*`, `/ajax-api*`, `/ar/*`, `/en/*`**	**H**
Salla App Store	`api.salla.dev/marketplace/v2/*`	Salla docs	**UNCLEAR**	No path disallows for `/marketplace/v2`	**L–M**
Aqar	`sa.aqar.fm/{category}/{city}` (RSC)	No English ToS found	**UNCLEAR (lean NO)**	**`Disallow: /graphql`, `/search/`, `/view/`, and used category slugs**	**H**

Syarah	`syarah.com/autos` (SSR)	No English ToS found	**UNCLEAR**	`Disallow: /api/`, `/*?*`; `/autos` allowed; two search endpoints explicitly allow-listed	**L–M**
Sabbar	`sabbar.com/ar/jobs`	No ToS found	**UNCLEAR**	**`User-Agent: *`** **`Allow: /`** (fully crawl-allowed)	**L**
Jarir (Constructor.io)	`ac.cnstrc.com/search/…`	Constructor terms not found; jarir.com robots	**UNCLEAR**	jarir.com disallows `/api/`, `/dp/*`; client hits a **separate** host	**M**
eXtra (Unbxd)	`search.unbxd.io/{apiKey}/{siteKey}/search`	Unbxd terms not found; extra.com robots	**UNCLEAR**	extra.com disallows `/search`; client hits a separate host	**M**
Etimad	`tenders.etimad.sa/Tender/AllSupplierTendersForVisitorAsync`	No ToS found (government platform)	**UNCLEAR**	`robots.txt` returns a **CAPTCHA / anti-bot challenge**	**M**
Kafiil	`kafiil.com/projects`, `/services`	No ToS found	**UNCLEAR**	Only register/login/purchase disallowed; `/projects`, `/services` **allowed**	**L–M**
Nice One	`niceonesa.com/ar/search?q=…`	No ToS found	**UNCLEAR (lean NO)**	**`Disallow: /search`**, `*?filters=*`, `*sort=*` — re-fetched, confirmed	**H**
Golden Scent	`goldenscent.com/ar/search?q=…`	No ToS found	**UNCLEAR (lean NO)**	**`User-Agent: *`** **`Disallow: /`** (full disallow) + `/search` — re-fetched, confirmed	**H**
Noon / LuLu	`noon.com/_svc/catalog/api/v3/u/search`; `gcc.luluhypermarket.com`	No ToS found	**UNCLEAR**	n/a	**M**

WAF-blocked / gracefully-degrading (documented in `DATA_EXPANSION_STUDY.md` §6): Noon, Aqar, Al-Dawaa,

Sharaf DG, Amazon, and Reddit `.json`. `MarketplaceRegistry.run_health_check()` isolates each client and labels failures `UNAVAILABLE` — that is an **availability** behavior, **not** a compliance mitigation.

3. Top compliance risks, ranked

1. **Reddit (all tiers)** — `Disallow: /` for `*`, plus a robots.txt pointer to "research and non-commercial use" and Data API Terms §3.1/§3.2 requiring a commercial agreement. The agent's purpose is commercial product design and the README explicitly claims to bypass Reddit's 403 walls. Clearest breach.
2. **Google Autocomplete + Google Trends RSS** — Google ToS bars automated access in violation of robots.txt and reverse engineering; Autocomplete is an undocumented internal endpoint (`client=firefox`) with no licence. Trends RSS has no commercial grant — the same structural gap that got Google News disabled.
3. **Robots-disallowed paths actually hit** — OpenSooq (`/api/*`, `/_next*`), Aqar (`/graphql` + category slugs), Nice One (`/search`), Golden Scent (`*` full disallow + `/search`). Concrete path-level violations, not gray areas.
4. **Arctic Shift** — redistributes Reddit content with no licence of its own; cannot cure Reddit's commercial-use restriction, and exists to evade Reddit's access controls.
5. **Reused public search keys** — Constructor.io / Unbxd / Algolia keys embedded in Jarir/eXtra frontends are driven from a separate Cloudflare Worker/backend for systematic catalog extraction. A search-only key is *designed to be public* for frontend use, but the retailer's vendor contract governs that key; third-party backend harvesting is a different use than the licensed on-site search UI, and the retail hosts' `robots.txt` disallow `/api/` and `/search`.

4. Recommended actions (research only — no code changes proposed)

- **Gate behind opt-in flags (like `GOOGLE_NEWS_RSS_ENABLED`), default OFF:** Google Autocomplete, Google Trends RSS, all Reddit tiers + Arctic Shift, Product Hunt feed, and the four robots-disallowed clients (OpenSooq, Aqar, Nice One, Golden Scent). Personal-experimentation-only until licensed.
- **Lowest-risk to keep** (still UNCLEAR commercial rights — keep behind a flag or add attribution/rate-limit discipline): HN Algolia API, Sabbar (`Allow: /`), Syarah `/autos` + allow-listed endpoints, Kafiil, Salla marketplace API, Haraj. **Do not** label any source "compliant" merely because no prohibition was found.
- **Partner / licence conversations:** Reddit (commercial Data API / Public Content Policy), Google (official APIs), Product Hunt (official OAuth2 API), and the retailers whose Constructor.io / Unbxd / Algolia keys are reused — or replace those with each vendor's own licensed API.
- **Operational:** have the health check respect `robots.txt` per host rather than only recording `UNAVAILABLE`; stop spoofing browser UAs for disallowed hosts.

5. Coverage & limitations

The full **Product Hunt legal text** (403), **Reddit's Public Content Policy body** (403 — relied on the robots.txt pointer plus the verified Data API Terms), and any published ToS/API terms for the **Arabic-language marketplace sites** (Mostaql/Khamsat terms are Arabic, transactional, silent on automated collection), **Constructor.io**, **Unbxd**, **Arctic Shift**, **Etimad** (CAPTCHA-gated robots) and the **Salla marketplace API** could not be retrieved — those entries are INFERRED / UNCLEAR, not cleared. Several targets are login-, WAF- or geo-gated, so live behaviour may differ from the fetched `robots.txt`. Google News RSS was already handled and excluded.

Independently re-fetched and confirmed by the coordinating agent: `www.reddit.com/robots.txt` (`Disallow: /`), `trends.google.com/robots.txt` (only `/explore?` blocked), `goldenscent.com/robots.txt` (`User-Agent: * Disallow: /`), `niceonesa.com/robots.txt` (`Disallow: /search`).

This audit covers source licensing only. It does not assess the project's own MIT LICENSE, which covers repository source code and grants no rights to third-party data.

Business Ideas from the Project's Own Findings

Date: 2026-11-11 **Classification:** a **spike** — an ideation/recommendation document, not code to keep and ship. No implementation, no code changes, no commits. **Method:** 4 parallel isolated research subagents (infrastructure data products, vertical B2B niches, consumer/demand-signal ideas, red-team validation) with self-contained briefs, then integration. Every high-stakes claim was independently re-verified against the repo by the coordinating agent. **Frameworks applied:** `research` (primary sources, single cited Markdown file), `market-gap-analysis` (competitor blind spots, demand-without-supply, underserved segments, price gaps), `evaluating-startup-ideas` (problem urgency / why-now / business math / validate-before-code), `idea-validation` (riskiest assumption, Wizard-of-Oz, pre-product invoice, fake door), `offers` (value equation + six-component offer anatomy), `dispatching-parallel-agents`.

⚠ Integrity note (this document obeys the guardrail it describes)

`signal_health.py:12-23` (`INTEGRITY_RULES`) forbids citing numbers that do not appear literally in source sections. We found that **both published reports violate this** — `"20,000+ daily searches"` (`reports/2026-11-10.md:14`), `"63% under 30"` (`LATEST_IDEAS.md:17`), `"300%-600%"` margins (`:20`), `">40%"` conversion (`:84`), `"cuaca besok" (expat influence)` (`reports/2026-11-10.md:16`), `"290,000+ tenders"` (`README.md:27`). Per `docs/CONSOLIDATED_STUDY.md:90-95`, every audited numeric claim is likely-inferred or unverifiable.

So in this document: - **No market sizes, TAMs, growth rates, customer counts or conversion rates are stated as fact.** - Every number is either **(a) quoted from a repo file with a `file:line` citation**, or **(b) marked ⚠ `ESTIMATE`** and used only as an illustration for the business math. - Idea scores are **judgment scores (1–10), not data**. - Banned sales vocabulary ("game-changing", "revolutionary", "10x", "secret", "worth $X", "100% guaranteed", fake "limited time") is deliberately avoided. - `docs/TOS_AUDIT.md:8` is the governing constraint: **zero data sources are declared "compliant" for commercial use.** `UNCLEAR` ≠ compliant.

Executive summary

The four studies converge on a single commercial insight: **this project's biggest liabilities and gaps are more sellable than its coverage.** It wired ~22 sources (`docs/CONSOLIDATED_STUDY.md:62`) and 13 clients (`:27`) before talking to a single buyer, audit shows zero commercial licences (`:60-71`), and the guardrail it built (`signal_health.py`) protects only future runs while both published reports contain unsourced numbers (`:88-95`).

Three ideas survive scrutiny. All three share two properties the project currently lacks: **they need no unlicensed scraped data to validate**, and **they can be sold manually before any code is written**. Everything else is blocked behind a licence conversation or rests on unverified evidence.

The most important conclusion is negative: **do not launch the current pipeline as a commercial product, do not add a 14th scraper, and do not build on the published reports' numbers.** The single missing primitive for the original vision is a cross-store EAN/GTIN resolver (`docs/CONSOLIDATED_STUDY.md:37,45`), not more coverage.

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1. Ranked idea portfolio

Scores are judgment (1–10) across **urgency** / **timing** / **viability** / **validation**, from the four subagents. "Validation" measures how cheaply demand can be proven — and it is the lowest column everywhere, which is the point.

Tier A — recommended, validation-first

#	Idea	Gap it sells	U	T	V	Val	Compliance posture

1	**Allowlist** — commercial-rights gate for AI agents	The ToS audit as a product	6	8	7	5	**No scraped data.** Reads `robots.txt` + published terms. Prototype needs no partnership.
2	**Award Radar (ترسية)** — price-to-win briefs for gov contractors	Etimad award results	9	7	6	3	Etimad visitor endpoints, `UNCLEAR`/Risk M + CAPTCHA (`docs/TOS_AUDIT.md:38`). Needs gov data licence.
3	**Claim Ledger** — provenance verifier for AI market reports	`signal_health.py` as a product	6	7	8	5	**Pure text processing** of the customer's own draft; MIT-licensed repo logic only. Zero third-party data.

Tier B — later, licence-gated or blocked

#	Idea	Why not now	U	T	V	Val
4	**Barcode Bridge** — GTIN + SAR price-history API	Highest asset value, but its identifiers live in `UNCLEAR` sources; needs a licence conversation first (`docs/CONSOLIDATED_STUDY.md:37,45`)	7	8	6	4
5	**Licensed Saudi SKU Feed** for BNPL/banks	Requires signed retailer data-sharing agreements; long cycles (`docs/TOS_AUDIT.md:58`)	7	8	5	3
6	**Salla App-Store Price & Complaint Brief**	Anonymous endpoint but "no standalone terms for bulk harvesting" (`docs/TOS_AUDIT.md:32`); buyers can eyeball it themselves	6	8	8	3
7	**District-Indicator Valuation Memo**	Sitemaps may not expose per-listing price/area; Aqar detail paths are HIGH-risk (`:33`)	7	7	5	3
8	**Offer-Clearing Salary Band** for HR firms	Sparse regex-only salary parse; `Allow: /` but "still no licence for commercial reuse" (`:35`)	6	6	7	3

Tier C — rejected (documented, not recommended)

The consumer/demand-signal agent was instructed to be skeptical and concluded the domain **is not worth pursuing**. Its reasoning is retained here because it is the most actionable negative finding:

Idea from `LATEST_IDEAS.md`	Why rejected	Score U/T/V/Val
ZATCA Arabic freelancer invoice kit	A downloadable template cannot emit Phase-2 signed XML/QR — product fails the regulation it sells against; `SYSTEM_PROMPT` forces consumer framing onto a merchant need	6 / 8 / 3 / 2
Istiraha / "جمعة" game-night pack	Incremental convenience, free Arabic printables exist; urgency rests on `20,000+` / `100,000+` unsourced figures	2 / 3 / 2 / 2
Bilingual ATS CV + RTL Notion OS	Crowded (self-scored 6/10 competition), "thousands of graduates"/`63%` unverified, free Canva/Notion substitutes	2 / 2 / 2 / 3
All 5 physical products (keycaps, matcha kit, kashta kit, e-incense, MagSafe mount)	Rest on `300%–600%` margins and `>40%` conversion — no source section carries them (`docs/CONSOLIDATED_STUDY.md:93`)	—

Rejected-list takeaway: the only defensible signal in this domain (ZATCA + freelance bid budgets) is a **merchant-compliance** need, not a consumer want. `docs/CONSOLIDATED_STUDY.md §4` says stop adding scrapers; the honest next step is 20 buyer interviews plus one paid 48-hour fake door, not another generated report.

2. Idea #1 in detail — Allowlist

The gap. `docs/CONSOLIDATED_STUDY.md:60–71`: ~22 wired sources, **zero declared "compliant"**, only Google

News RSS actually gated (`config.py:19-21`; `agent.py:500`), 7 HIGH-risk sources. `docs/TOS_AUDIT.md §4` recommends gating ~10 more; enforcing it leaves ~27% of the daily pipeline (`:73`). `signal_health.py` records `UNAVAILABLE` but does **not** enforce `robots.txt` (`:44,59`). Competitor blind spot: `robots.txt` parsers check *syntax*, not *commercial-use licensing*. Pain-point evidence: this repo built 22 sources *before* auditing terms (`docs/CONSOLIDATED_STUDY.md:79-80`). Underserved segment: small agent/RAG shops with no counsel. Price gap: counsel review vs. an automated gate.

Customer & problem. Teams shipping scraper or RAG agents hit procurement blocks or takedowns when a source forbids commercial use — Reddit's Data API Terms §3.1/3.2 require a commercial agreement (`docs/TOS_AUDIT.md:24`), Google News RSS is personal-only (`config.py:13-16`). Hair-on-fire at a cease-and-desist or enterprise security review; otherwise a nice-to-have. This is the unglamorous-niche play (compliance plumbing), not consumer social.

Why now — with the red team's caveat. Agents moving prototype→production is a plausible inflection, but the red team found **no dated, primary-source-verified external inflection in this repo**: ZATCA Phase-2 appears only as a Zid feature (`CANDIDATE_MARKETPLACES_FLEET.md:61`) and a merchant pain point (`reports/2026-11-10.md:20`); the only primary link is a *VAT* guideline (`DATA_EXPANSION_STUDY.md:94,131`). Genuine headwinds do exist — PA-API 5 is dead and X's free tier ended Feb 2026 (`:72,82,95`). **Do not build a sales narrative on an unverified deadline.**

What you sell (autonomous outcome). `URL → allow / deny / needs-licence, with the governing clause cited`, enforced as a CI/ingest gate plus an audit trail. A verdict, not a dashboard.

Assets + compliance. Reads `robots.txt` and published ToS/terms URLs (legal text, not scraped catalog data). Zero dependence on the 7 HIGH-risk sources — it *classifies* them. The seed dataset is the repo's own `docs/TOS_AUDIT.md`. Partnership: counsel review of verdicts; `docs/TOS_AUDIT.md:8` states the audit is research, not legal advice.

Business math. ⚠ `ESTIMATE / illustration only — not verified`: ⚠ SAR 24k ACV × ⚠ 40 teams = ⚠ SAR 960k ARR. No comparable, no buyer evidence in the repo (`docs/TOS_AUDIT.md:6`: "a side project with no revenue").

Riskiest assumption. Teams will pay to *disable their own scrapers* instead of ignoring the risk.

Cheapest validation. Manual Vanta-style service: hand-audit 10 agent pipelines, deliver a "sources to gate" memo, ask for a **paid retainer before any code**. The schlep is reading terms and selling — exactly the non-coding work every study says was skipped (`docs/CONSOLIDATED_STUDY.md:50,111`).

Top risk. Verdicts are `UNCLEAR`, not legal clearance; buyers may not pay for hedged answers.

Six-component offer (per `offers`)

#	Component	Design
1	**Core deliverable**	A written Source Rights Ledger for one agent pipeline: every outbound host, the governing clause URL, and an allow / deny / needs-licence verdict; plus a drop-in CI gate config. **Includes:** up to ⚠ 25 hosts, one re-audit within 30 days. **Doesn't include:** legal opinion, licence negotiation, monitoring after delivery.
2	**Bonus stack**	(a) Starter ledger pre-filled from `docs/TOS_AUDIT.md` for the 7 HIGH-risk Saudi sources → closes "we have no counsel"; (b) one-page stakeholder memo for the security review → closes "my team won't adopt"; (c) 30-day async Q&A → closes "we'll get stuck". Additive, no inflated "value" claims.
3	**Guarantee**	**Paid-pilot, conditional exit** (the `guarantee-design.md` productized-service pattern): "First engagement is a paid pilot. If the ledger doesn't cover every host you actually call, you don't pay the second half and we end on good terms." Measurable, no over-promising, no "100% guaranteed".
4	**Scarcity / urgency**	**Real capacity only**: manual delivery limited to ⚠ a small number of pilots per month because each audit is hand-read. No countdown, no fake deadline — the repo has no verified external deadline to anchor one.
5	**Name**	**Allowlist — Source Rights Pilot**. Specific, describes the verdict, no hype adjectives.

6	**Price + payment**	⚠ ESTIMATE: ⚠ SAR 4,500 paid pilot (50% on booking, 50% on delivery), then ⚠ SAR 2,000/month retainer for quarterly re-audits. Paid in SAR; Mada / bank transfer. All figures unverified assumptions, not data.

Value-equation diagnosis (`offers/references/value-equation.md`): dream outcome 6, **perceived likelihood 4 (binding constraint)**, time delay 8, effort 7. The weak lever is *perceived likelihood*, because verdicts are `UNCLEAR` rather than legal clearance. Fix it with proof (named pipelines audited, clause-level citations, the paid-pilot guarantee) — **not** by lowering price or adding bonuses. A one-component change targets a 10–40% relative lift; multiply, don't add, and confirm with a test.

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3. Idea #2 — Award Radar (conditional on a one-request spike)

The gap. Etimad is wired as an anonymous visitor client (`marketplace_registry.py:48`) but captures only live RFP metadata. `DATA_EXPANSION_STUDY.md:109` calls award results (winning bidder, all bidder prices, awarded value) the "single highest-leverage new dataset" — **and explicitly marks it INFERRED, not live-probed** (re-verified: that exact wording is in the file). `README.md:27`'s "290,000+ tenders" is unverified. Blind spot: contractors see open RFPs, never settled outcomes.

Customer & why now. Mid-size facility-management / IT / civil subcontractors in one region. Bid preparation costs money; a lost bid with no price-to-win feedback is a recurring quantified loss. Genuine hair-on-fire (urgency 9) — but the "why now" rests on Vision 2030 procurement and ZATCA, neither of which has a dated primary source in the repo.

What you sell. A weekly per-contractor "bid / no-bid + price-to-win" memo: last N awards for their agency + activity + region, winning bidder, every bidder's price, awarded value, recommended bid band. A result, not a dashboard.

Compliance. `docs/TOS_AUDIT.md:38`: `UNCLEAR`, Risk M, `robots.txt` returns a CAPTCHA anti-bot challenge, no published ToS. Needs a government-data licence via the Etimad developer portal (`DATA_EXPANSION_STUDY.md:133`) and counsel review.

Business math. ⚠ ESTIMATE: ⚠ SAR 18,000 × ⚠ 25 firms = ⚠ SAR 450,000 ARR. Illustration only; no buyer evidence.

Riskiest assumption (kill it first). The award endpoint actually returns winning bidder + all bidder prices at usable coverage. **Cheapest experiment:** one read-only GET to confirm the payload, *then* a manual Wizard-of-Oz PDF for one agency/region sent to 20 contractors with a pre-product invoice ask — before any ingestion code.

Top risk. CAPTCHA gate + zero commercial licence; supply can be withdrawn. Start the licence conversation on day one.

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4. Idea #3 — Claim Ledger

The gap. `signal_health.py:12-23` (`INTEGRITY_RULES`) + `:52-60` (`health_block`) are injected at `agent.py:544`, yet both published reports contain **0** `UNAVAILABLE` tokens and unsourced numbers (`docs/CONSOLIDATED_STUDY.md:88-94`). LLM eval tools grade fluency, not claim-to-source lineage.

What you sell. `Draft in → annotated report out`, with every unsourced numeric claim flagged or replaced with "insufficient data". A verified artifact, not a dashboard.

Why it is unusually clean. Pure text processing of the customer's own draft + their own source sections; MIT-licensed repo logic only (`LICENSE`; `docs/TOS_AUDIT.md:69`). No scraping, **zero dependence on the 7 HIGH-risk sources**, no partnership needed. Highest viability score (8).

Validation. Wizard-of-Oz: take this project's own 3 published reports, hand-produce a claim ledger exposing the ~8 unsourced numbers (`docs/CONSOLIDATED_STUDY.md:90-94`), show to 5 research leads, ask for a **paid pilot** before code.

Top risk. Buyers treat fabricated stats as insufficiently urgent to pay per report, or say "our LLM is fine".

5. Anti-ideas — do not do these

From the red-team agent, each with the finding that kills it:

1. **Launch the current ~22-source commercial data product.** `docs/TOS_AUDIT.md:8,57` clears zero sources; `LICENSE` covers repo code only, **no third-party data rights** (`:69`). Legal posture is already commercial (`:6`).
2. **Add a 14th scraper.** `DATA_EXPANSION_STUDY.md:13`: "It has a storage / identity problem." `docs/CONSOLIDATED_STUDY.md:37`: "A 14th scraper adds another incomparable price list, not a comparison."
3. **Chase Al-Dawaa / Namshi / Alsaif first on fleet scores.** Al-Dawaa ranked #1 (8.2, `CANDIDATE_MARKETPLACES_FLEET.md:224`) but a live Cloudflare 403 killed it (`DATA_EXPANSION_STUDY.md:32,93`); the 1 built platform (LuLu 6.8) scores *below* the unbuilt 7.195 average (`docs/CONSOLIDATED_STUDY.md:52-56`).
4. **Monetise Reddit / Arctic Shift data.** `Disallow: /`, personal-only, Data API Terms §3.1/3.2 (`docs/TOS_AUDIT.md:24-25`), and `README.md:15` is a written admission of "bypassing Reddit's 403 rate-limiting walls".
5. **Publish AI market reports with the current provenance.** `INTEGRITY_RULES` protects future runs only (`docs/CONSOLIDATED_STUDY.md:86,103`); `README.md:67` describes behaviour the shipped reports do not exhibit.
6. **Build consumer products on the reports' numbers.** All the figures catalogued in the Integrity note above are likely-inferred or unverifiable (`:90-95`).
7. **Sell a price-comparison product before the EAN resolver.** `/api/compare` never matches (`cloudflare-worker/src/index.ts:67-99`); LuLu drops `ean` (`lulu_client.py:85-90`); `Env` empty (`:12-14`).

6. Five riskiest assumptions, ranked

#	Assumption	Evidence	Cheapest falsifying experiment
1	**A buyer will pay.**	No customers, no revenue, no demand validation; 13 clients + 20-platform fleet built pre-buyer (`docs/CONSOLIDATED_STUDY.md:27`). 27 tests stub all network (`:98-101`).	**Pre-product invoice** to 20 cold prospects. Pass = one paid invoice from a stranger. Fail = "sounds useful".
2	**The data can be used commercially.**	Zero compliant sources (`README.md:77`, `docs/TOS_AUDIT.md:8,57`); MIT grants no data rights (`:69`).	Written commercial-use confirmation from one §4 "keep" source. Silence or "no" falsifies it.
3	**The mined signals are true.**	0 health markers, unsourced numbers, guardrail future-only (`:88-95,103`).	**Wizard-of-Oz**: hand-build one brief, trace every number to a literal raw line. Fail = any invented number.
4	**Cross-store matching is solvable.**	No resolver; `ean` dropped; `/api/compare` is not a comparison (`:35,45`).	Manually-resolved CSV across the 3 Worker retailers, matching by EAN/MPN by hand. Fail = most rows have no shared key.
5	**Coverage is the moat.**	Constraint moved to storage + identity (`DATA_EXPANSION_STUDY.md:13`); WAF/partner ceilings (`:68,122`).	Fake door with a real price, zero scrapers. Fail = no cold signups.

7. Compliance decision rule

- **Use today:** only the repo's own MIT code (`LICENSE`). No third-party data qualifies. Google News RSS is personal/non-commercial only and gated off (`config.py:19-21`).
- **Licence / partnership first (default OFF):** the six "lowest-risk to keep" — HN Algolia, Sabbar, Syarah, Kafiil, Salla marketplace, Haraj — are still `UNCLEAR`, not licensed (`docs/TOS_AUDIT.md:57`). Plus written grants from

Reddit, Google, Product Hunt and the retailers whose Constructor.io/Unbxd/Algolia keys are reused from a separate backend (`:52,58`).

- **Must drop:** the 7 HIGH-risk — Google Autocomplete, Reddit RSS + Arctic Shift, OpenSooq, Aqar, Nice One, Golden Scent (`docs/CONSOLIDATED_STUDY.md:66`; `docs/TOS_AUDIT.md:48-51,56`). Stop UA spoofing; make health checks respect `robots.txt` (`:59`).
- Enforcing §4 leaves ~27% of the pipeline / ~46% of clients (`docs/CONSOLIDATED_STUDY.md:73`), all still `UNCLEAR`. Confirm with counsel (`docs/TOS_AUDIT.md:8`).

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8. Validation-first plan — no code until demand is proven

Step	Artefact	Pass / fail signal
1	**20 cold-customer interview notes**: what do you already pay for / do manually to track prices or demand?	Pass = repeated, specific, already-paid-for workflow. Fail = polite interest → **STOP**.
2	**One-page pre-product invoice** for a hand-made weekly brief at a stated SAR price.	Pass = ≥1 paid invoice / deposit from a cold contact (financial commitment). Fail = zero → **STOP**.
3	**Manually-resolved CSV + Wizard-of-Oz week-one brief** from the six §4 sources, every number traced.	Pass = buyer asks for week 2 AND all figures trace to raw data. Fail = untraceable numbers → **STOP**.
4	**Written rights pack + fake-door page** with a real price, for exactly the sources Step 3 used.	Pass = written commercial grants + cold inbound. Fail = silence → shrink to the granted set. **Only now write code** — starting with the EAN resolver (`docs/CONSOLIDATED_STUDY.md:118`), never another scraper.

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9. Concrete next steps (prioritised)

1. **Pick one idea and run Step 1–2 of §8 this week.** Recommended: **Allowlist**, because it needs no unlicensed data and its validation is a paid manual audit — the exact discipline the project skipped.
2. **Kill Idea #2's riskiest assumption with one read-only GET** to the Etimad award endpoint before any further work on it. It is explicitly `INFERRED` (`DATA_EXPANSION_STUDY.md:109`).
3. **Produce a Claim Ledger for the project's own 3 published reports** as a free internal artefact — it doubles as the proof asset for Idea #3.
4. **Regenerate or clearly label the pre-guardrail reports**, or fix `README.md:67`, which currently claims behaviour the shipped reports do not exhibit.
5. **Only after a paid invoice**: build the EAN/GTIN resolver, starting by extracting LuLu's `ean` (`lulu_client.py:85-90`). Do not add a 14th scraper.

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10. Limitations & boundaries

- **Compliance verdicts are inherited** from `docs/TOS_AUDIT.md` (research, not legal advice). No new ToS/robots research was performed in this pass; `UNCLEAR` does not mean compliant.
- **All business math is ⚠ `ESTIMATE`.** The repo contains no customer, revenue, ACV, conversion or market-size evidence (`docs/TOS_AUDIT.md:6`). Score columns are judgment, not measurements.
- **The consumer-domain rejection is a reasoned judgment**, not proof there is no market; it is based on the absence of verifiable demand signals in the repo and the documented fabrication problem.
- **Idea #2's core dataset is INFERRED**, flagged in its source study (`DATA_EXPANSION_STUDY.md:109`) — it is included precisely because one GET can falsify it cheaply.
- **Why-now claims are weak**: the red team found no dated, primary-source-verified external inflection in the repo;

several candidate inflections are headwinds (PA-API 5 dead, X free tier ended Feb 2026).

- **This is a spike / recommendation document.** No code was written, no tests added, no files outside `docs/BUSINESS_IDEAS.md` were changed, no commits or pushes. Existing working-tree changes remain uncommitted.
- The four subagents were told not to write files; their outputs were returned as reports and integrated here. High-stakes claims (Etimad INFERRED wording, `signal_health.py` guardrail lines, `LICENSE`, skill reference files, unverified report figures) were independently re-verified by the coordinating agent.

Website Feasibility Study — Arabic / Saudi AI-Content Cleanup

Date: 2026-11-11 **Classification:** a **spike** — a feasibility question whose output is a recommendation, not code to keep or ship. No implementation, no code changes, no commits. **Question:** *Is it worth opening a website targeting the Arabic/Saudi "AI text cleanup" problem, using all the APIs and data we have — and what should it be called?* **Method:** full verification sweep; 3 parallel isolated research subagents (data/API inventory, demand+feasibility, naming) with self-contained briefs; live read-only probes of the repo's **safe** public APIs; independent re-verification of quantitative claims by the coordinating agent. **Frameworks:** `research` (primary sources, single cited Markdown file), `dispatching-parallel-agents`, `brainstorming` (spike path), `evaluating-startup-ideas`, `idea-validation`, `offers`.

⚠ Integrity note

This document obeys the guardrail it describes (`signal_health.py:12-23 INTEGRITY_RULES`). **No market sizes, TAMs, growth rates, conversion rates or buyer counts are stated as fact.** Every number is either (a) computed from repo files with the command shown, (b) quoted from a primary source with a URL, or (c) marked ⚠ `ESTIMATE / unverified`. Banned sales vocabulary is avoided. `UNCLEAR` ≠ compliant (`docs/TOS_AUDIT.md:8`).

Correction to a prior answer: last turn I quoted a "Fractl survey" (8-in-10 workers, median 45 min cleanup, 62% redid a colleague's work, 23% reached customers). Fresh verification found that attribution **unreliable** — those figures trace to *workslop* coverage, not a Fractl survey, and could not be confirmed against a primary source. **They are withdrawn and must not be used in a pitch.** `docs/CONSOLIDATED_STUDY.md:88-95` documents exactly this failure mode in the repo's own published reports.

Executive summary

Question	Answer
Is a full website / SaaS product worth it?	**NO.** Not on directional platform statistics, with zero buyer evidence, against a $5 commodity floor and free humanizer tools.
Is a one-page site worth it?	**YES, conditionally** — strictly as a time-boxed **validation instrument** (a fake door with a real price), not as a product.
Does "we have a lot of data" help?	**NO.** It is ~1.7 MB of legally uncleared Saudi retail listings that this product neither needs nor may lawfully use.
Recommended name	تَنْقِيح | **Tanqih** — "source-checked revision". Tagline: «نردّ كل رقم وادعاء إلى مصدره، ونسلّمك النص مُعلَّماً مع سجل التغييرات.»

The single most important finding is negative but useful: **the customer's-own-document model sidesteps the entire `docs/TOS_AUDIT.md` problem.** The audit concerns licensing of third-party data the project harvests; a text-cleanup service that only processes a document a client voluntarily sends has no source-licensing question at all.

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1. What the APIs and data actually gave us

Three agents were asked to use the repo's assets. Here is what came back, with verification.

1.1 Local data inventory (re-verified by the coordinating agent)

Command: `python verify_counts.py` (counts files, bytes, JSON top-level list items, SQLite rows, Python lines).

Measure	Verified value
Top-level `.json` files	**8**
Total JSON bytes	**1,824,995 bytes ≈ 1.74 MiB**
Top-level list items across those files	**834** (`haraj_100_gaming_listings` 100 + `jarir_taxonomy_tree` 12 + `stream_data` 722)
SQLite rows (`haraj_market.db`)	**154** (85 `listings` + 69 `price_history`)
Python files / lines	**102** / **8,176**

So "a lot of data" resolves to ≈ **988 structured records** (834 + 154, *INFERRED* record definition) plus ~1.7 MB of raw harvest dumps, GraphQL introspection and category trees. Three of the four published reports are the **same file** byte-identical (`LATEST_IDEAS.md`, `docs/LATEST_IDEAS.md`, `reports/2026-11-11.md` — md5 `b16cb77b52e8`, 42,624 B). The bulk of the repo is code, recon scripts, downloaded JS/HTML bundles and AI-written prose — **not a cleaned dataset**.

1.2 Live read-only API probes (safe sources only)

Deliberately did **not** touch the 7 HIGH-risk sources (`docs/TOS_AUDIT.md §4`). Command: `python probe_apis.py`.

API	Result	Reading
Amazon SA autocomplete (`DATA_EXPANSION_STUDY.md:25`, VERIFIED 200, no key)	محتوى ذكاء اصطناعي → []; تدقيق نصوص → []; تصحيح محتوى → []	**Zero Arabic suggestions.** A negative demand signal from this API — but it is a *product* search, so this is weak evidence, not proof of no demand.
HN Algolia (`docs/TOS_AUDIT.md:28`, risk L)	`AI content audit` nbHits **2,789**; `AI slop cleanup` **638**; `AI humanizer` **498**; `hallucination editing` **76**	Counts of HN stories matching a query — a **weak topical-attention proxy**, NOT market demand or buyers. Do not present as demand.
Salla App Store (repo `salla_client.py`)	**19 categories**, including a first-class الذكاء الاصطناعي category. q=4 → تدقيق apps (collections, SEO, ops, reporting); q=15 → محتوى apps (SEO, marketing, loyalty, UGC)	**No AI-content-audit app exists in the merchant app store.** A genuine whitespace signal — but it is an *app store for merchant tools*, not a service marketplace, and "no app" ≠ "no demand".

1.3 Blunt verdict on the data

`docs/TOS_AUDIT.md:8` — **zero sources declared "compliant"**; §4:57 "Do not label any source 'compliant'"; `README.md:77` says so. The MIT `LICENSE` covers repo code only (`docs/TOS_AUDIT.md:69`), granting no third-party data rights. A website built on this scraped data would inherit those rights problems wholesale; `docs/BUSINESS_IDEAS.md:141` already lists "launch the ~22-source commercial data product" as an **anti-idea**.

Technically it is also the wrong data: there is no corpus of Arabic work documents, no AI-tell labels, no tone annotations, no customer records, no Arabic NLP (`requirements.txt` = `google-antigravity`, `httpx`, `python-dotenv`, `beautifulsoup4`; no `camel-tools` / `farasa` / `hazm` / `transformers`). The only transferable asset is the **method**: `signal_health.py:12-23 INTEGRITY_RULES` (forbid invented numbers/terms, require "insufficient data", literal citation) — pure text, network-free, MIT-licensed, zero third-party data. That is a discipline, not a dataset.

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2. Demand and competition (fresh verification)

Claim	Source	Verdict

Freelancer.com "correct AI / hallucination / error" listings **+87% → 10,760** (Aug 2025–Jun 2026); Upwork remediation gigs **+70% YoY**; Fiverr "AI cleanup" searches **>20×** since 2023	Guardian, Search Engine Journal	**VERIFIED AS REPORTED / DIRECTIONAL** — platform-internal data, explicitly unaudited; the three platforms measure different things and cannot be summed
Four **PwC Middle East** reports with fake footnotes, incl. a hallucinated paper on *air quality in Riyadh* (Jul 2026)	FT, GPTZero	**VERIFIED** — the strongest KSA-specific proof point for a sales conversation
Arabic ≈ **0.5% of web content** (Qatar's Fanar researchers); Saudi/Gulf dialect + register failures; "unlived Arabic"	Fast Company ME	**VERIFIED** — the technical moat is real
"A fast draft is not publish-ready content"; tone before grammar; **PDPL** warning on pasting customer data into public AI tools	Origami.sa (Saudi agency, 11 Jun 2026)	**VERIFIED**
Ipsos/ABG UAE+KSA: **70% concerned** about overreliance without human oversight	Ipsos	**VERIFIED (70%)**; but "112 professionals / 68% / 2026" is **UNVERIFIED** — the page says "100+", 96% used AI tools, dated 30.06.**2025**. ⚠ Survey, not buyer counts
Deloitte "Arab Media in the Age of AI": provenance + human-in-the-loop	Deloitte	**PARTIAL / UNVERIFIED** — the quoted "provenance" wording and "KSA 70%" are not on the public page. Forecast, not buyer counts
The exact service is already sold in Arabic	Khamsat ($5 / 400–500 words), Kafiil ($5), Mostaql (buyer wrote "عدم استخدام الذكاء الاصطناعي", 50–$25)	**VERIFIED** — real demand, but at commodity prices
NEW: Khamsat now lists "**AI** تدقيق محتوى" as a category; a buyer request asks for "مراجعة بشرية… أنسنة المحتوى"	`khamsat.com`	**VERIFIED** — the category exists, i.e. demand is organized, not a lone gig
NEW: Arabic AI humanizer/detector SaaS already exists (`aidetector.pro`, `humanifyer.com`, `gptcleanup.com`, Undetectable.ai)	as listed	**VERIFIED** — crowded, and several sell detector *bypass*
Arabic AI-tell word lists are weakly sourced	`content-locale-humanize-ar` skill ("sourcing confidence: the weakest of any language checked")	**VERIFIED** — do not build an automated "AI tell detector"
Disclosure pressure: SDAIA guidance, Google Ads AI-labelling (Jul 2026), Meta/TikTok	Google Ads policy, Eclipse Agency	**VERIFIED** — sell auditing, never hiding

Counter-signal (same Guardian piece): freelancers expect humanizing work to "dry up within five to 10 years as models improve". Demand is real but possibly transient. And the Guardian reports clients **lowball** cleanup work.

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3. Verdict on the website

(a) Full product / SaaS site — NO

Conditions to reconsider *only*: ≥1 paid invoice from a cold stranger (`docs/BUSINESS_IDEAS.md:155,177`), written data-rights clearance, and a defensible price above the ~$5 commodity floor.

Reasons: 1. `docs/CONSOLIDATED_STUDY.md:27` — 13 clients and a 20-platform research fleet were built **before any**

buyer conversation; `docs/TOS_AUDIT.md:6` — no revenue. A full site repeats that exact mistake. 2. Demand stats are platform-internal and directional, not buyer counts. 3. The service is already commoditized (Khamsat category, $5 gigs, free humanizer SaaS). 4. The only real moat — native Saudi/Gulf register — requires a **human editor**, which is a service, not scalable software (`content-locale-humanize-ar` confirms Arabic tell-lists are unreliable). 5. `docs/BUSINESS_IDEAS.md:29`: the most important conclusion is negative — don't launch on unverified numbers.

(b) One-page fake door / lead-gen site — YES, conditionally

Endorsed by the repo itself: `docs/BUSINESS_IDEAS.md:66` ("one paid 48-hour fake door"), `:159` ("fake door with a real price, zero scrapers"), `:179` (Step 4). Per `idea-validation`, validate demand **before** code; the site is a **measuring device, not a product**.

Conditions: a real stated SAR price; no accounts, no upload storage, no AI processing, no scrapers; the CTA must ask for **money** (paid pilot deposit / pre-product invoice), not an email; pre-declared pass/fail and a time box. **A page with no traffic is a null experiment** — pair it with 20 cold outbound contacts (`docs/BUSINESS_IDEAS.md:176`).

Riskiest assumption and cheapest falsification

Assumption: a KSA/GCC buyer will pay a foreign, no-local-entity provider a premium to clean up their *own* Arabic work documents, repeatedly, instead of a $5 Khamsat/Kafiil gig, a free humanizer, their own LLM, or an in-house editor. This bundles willingness-to-pay, cross-border trust and preference over cheaper substitutes — the repo's #1 ranked assumption (`docs/BUSINESS_IDEAS.md:155`).

Cheapest experiment: 48-hour one-page fake door at a real price + 20 cold outbound contacts + a pre-product invoice/deposit ask for **one hand-edited Wizard-of-Oz document**. **Pass** = ≥1 paid deposit from a stranger. **Fail** = polite interest, email signups or silence → stop. No code, no accounts, no backend.

Business math — ⚠ all ESTIMATE

⚠ `ESTIMATE / unverified`: ⚠ SAR 300 per document audit × ⚠ 2 documents/month × ⚠ 12 months = ⚠ SAR 7,200 ACV × ⚠ 20 retained clients = ⚠ SAR 144,000/yr. **Every input is invented; this is arithmetic, not a forecast.** The repo has zero buyer evidence (`docs/TOS_AUDIT.md:6`). The ⚠ SAR 300 price is the weakest input, given $5 competitors.

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4. Compliance / decision rule for the website

May touch: only the customer's own document, voluntarily supplied with explicit consent, processed transiently, then deleted.

May not touch: any scraped third-party data — marketplace listings, Reddit / Google Trends / Autocomplete, retailer catalogs, any of the ~22 sources in `docs/TOS_AUDIT.md`.

A customer-owned-document model avoids the audit problem entirely. `docs/TOS_AUDIT.md` concerns licensing of data the project *harvests from third parties*; with no third-party ingestion, the HIGH / UNCLEAR verdicts are irrelevant. This is the same logic as "Claim Ledger": pure text processing of the customer's own draft (`docs/BUSINESS_IDEAS.md:43,129`).

PDPL: if a document holds personal data, Saudi PDPL (SDAIA) applies; Origami.sa warns pasting customer data into public AI tools may violate it. The operator is outside KSA, so cross-border processing is an open question — **confirm with counsel**, not asserted here (`docs/TOS_AUDIT.md:8` is research, not legal advice). **Disclosure:** SDAIA guidance and platform labelling rules push toward disclosure; the service must support it, never evade it.

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5. Day-one scope (YAGNI) and prohibited claims

Does: one static page with (1) one specific service — "human review of Arabic work documents: invented facts/numbers, wrong Saudi/Gulf register, AI tells"; (2) one real SAR price, one deliverable (annotated doc + change log), one CTA "request a paid pilot"; (3) a plain privacy/disclosure note (what data, where processed, retention/deletion,

no third-party AI tools on your data, "we do not help hide AI use"); (4) email/WhatsApp contact. **No** accounts, login, stored uploads, payment gateway, dashboard, API, detector, scraper or AI processing. Invoice manually.

Must NOT claim: "AI detector", "% AI-written" or any AI score (Arabic detection unreliable); "bypass / undetectable" or any evasion of AI-use disclosure; "100% guaranteed" (use the paid-pilot conditional exit, `docs/BUSINESS_IDEAS.md:96`); any market size / growth / conversion figure as fact; any KSA presence, local entity or local address; any banned sales vocabulary.

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6. Name: تَنْقِيح | Tanqih

Recommended. Arabic تنقيح = critical revision / refinement of a text (تنقيح النصوص = textual verification). It names the **method**, not the machine; it is one short word, spells unambiguously in Latin, and an obvious-brand screen surfaced no clear collision (⚠ UNVERIFIED, not trademark clearance).

- **Arabic tagline:** «نردّ كل رقم وادعاء إلى مصدره، ونسلّمك النص مُعلَّماً مع سجل التغييرات.»
- **English tagline:** "Every figure and claim traced to its source — delivered as a marked-up redline and a change log."

Runner-up سَنَد / **Sanad** ("documentary evidence") is semantically perfect but the screen found 6+ active Saudi/Gulf "Sanad" brands, two finance entities, Jordan's `sanad.gov.jo` government digital-ID platform and Al Jazeera's Sanad fact-check unit — a trademark risk and a direct hit on the "don't be confused with a government body" constraint.

Rejected: ميزان / Mizan (suspected Meezan Bank + Quranic authority overtones), رصد / Rasd (surveillance/detection register → drifts toward AI-detector framing), تدقيق / Tadqiq (the generic category word buyers already use — cannot be a named offer).

English alternates (use only on the foreign-entity surface): *Sourcebound* (international landing page / email domain), *Redline Audit* (named offer on proposals/invoices), *Claim Ledger* (deliverable title). Saudi-facing copy and Khamsat/Kafiil/Mostaql gig titles stay تنقيح.

Positioning caveat: copy saying «إزالة آثار الذكاء الاصطناعي / إخفاء / تجاوز الكشف / نسبة التوليد» attracts buyers seeking detector evasion. Put the boundary in the FAQ **and** onboarding: «لا نتحقق من نسبة التوليد ولا نساعد على إخفاء استخدام الذكاء الاصطناعي؛ نُدقّق الأرقام والادعاءات ونُوثّق كل تعديل وسببه.» Sell the audit, the redline and the log — never the disappearance.

Pre-commit checklist (all UNVERIFIED — none run): (1) domains `tanqih` across .com/.sa/.com.sa/.co + misspellings + Arabic IDN; (2) SAIP trademark search (Nice 35/41/42) **and** Ministry of Commerce trade-name reservation — these are different; (3) authority collision with SDAIA / ZATCA / MCIT / .gov.sa; (4) seller names, gig titles and slugs on Khamsat, Kafiil, Mostaql, Ureed; (5) handles on X (primary Saudi B2B channel), LinkedIn, Instagram; (6) 2–3 native-Saudi readers say the name + tagline aloud and answer unprompted "what does this do?" — check for Egyptian/Levantine read, stiff literary MSA, religious/legal authority, or "AI detector" read.

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7. Validation-first sequence (no code until a stranger pays)

Step	Artefact	Pass / fail
1	**20 cold-customer interview notes** — what do you already pay to fix in Arabic work documents?	Pass = repeated, specific, already-paid-for workflow. Fail = polite interest → STOP
2	**One-page fake door** named تنقيح, real SAR price, "request a paid pilot" CTA	Pass = ≥1 paid deposit from a cold stranger. Fail = email signups / silence → STOP
3	**One hand-edited Wizard-of-Oz audit** of a real client document — redline + change log, every number traced	Pass = buyer asks for week 2 AND all figures trace to source. Fail = untraceable claims → STOP
4	**Written rights + disclosure pack** for exactly the data Step 3 used	Pass = written consent + no third-party ingestion. **Only now write code**

8. Limitations & boundaries

- **Compliance verdicts are inherited** from `docs/TOS_AUDIT.md` (research, not legal advice). No new ToS review here.
- **Demand figures are platform-internal, directional and unaudited**; the withdrawn "Fractl" figures must not be reused. HN Algolia `nbHits` and Amazon autocomplete results are weak topical proxies, **not** demand data.
- **All business math is ⚠ ESTIMATE.** Zero customers, invoices, ACV or conversion evidence exists in the repo (`docs/TOS_AUDIT.md:6`).
- **"No AI-content-audit app in Salla App Store" is a whitespace signal, not proof of demand** — an app store for merchant tools ≠ a service marketplace.
- **The name is not cleared.** Domain, SAIP/MoC, board, handle and native-Saudi register checks are all UNVERIFIED and are the user's to run.
- **Arabic fluency is the deciding fork.** The core deliverable requires native/near-native Arabic with Saudi register sensitivity (`Alignerr` Arabic-language-expert role, $40–120/hr remote, requires exactly that). If the operator lacks it, the model becomes an agency with a subcontracted editor — a different, harder business with thinner margin. This study does not assume that question is answered.
- **Why-now is weak:** the red-team study found no dated, primary-source-verified external inflection in the repo; several candidates are headwinds (PA-API 5 dead, X free tier ended Feb 2026).
- **This is a spike / recommendation.** No code, tests, files outside `docs/WEBSITE_FEASIBILITY_STUDY.md`, commits or pushes. Live API probes were read-only, on safe sources only, and touched none of the 7 HIGH-risk sources.